A zero-free half-plane for the Riemann zeta function: $\zeta(s) \neq 0$ for $\operatorname{Re} s > 23/24$

An exposition of OpenAI's *The Quasi-Riemann Hypothesis*, matched to a Lean formalization

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Abstract

We give an account of a proof that the Riemann zeta function has no zeros in the half-plane $\operatorname{Re} s > 23/24$. The argument is that of OpenAI's *The Quasi-Riemann Hypothesis*, which proves the half-plane $\operatorname{Re} s > 11/12$ for all Dirichlet L-functions and all finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$, specialized here to ζ and to a fixed value of its parameter. A Möbius sum over the ideals of $\mathbb{Z}[\omega]$ is embedded in a sextic-twisted family. Poisson summation converts the family's mean square into one with cubic Gauss-sum coefficients; Kubota's cubic theta function converts the sextic twists into quadratic ones; the quadratic large sieve then bounds the completed sums, and a finite descent removes the cube factors. The argument is matched to a Lean 4 formalization: every result of Section 3 is a Lean theorem with the same statement, and the main results of Sections 4–7 are theorems of the formalization, proved with somewhat different bookkeeping.

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1 Introduction

The *quasi-Riemann hypothesis* for $\zeta(s)$ is the assertion that there is a fixed $\delta > 0$ such that $\zeta(s)$ has no zeros in the half-plane $\operatorname{Re} s > 1 - \delta$ (e.g., [1, §1], [20, p. 274], [2, §2]). The classical zero-free regions of de la Vallée Poussin [26] and of Vinogradov and Korobov [27, 15] shrink to

the line $\operatorname{Re} s = 1$ as the height grows; a half-plane of fixed width is a different kind of statement. This note gives an account of one proof.

Theorem 1.1. *The Riemann zeta function has no zeros in the half-plane* $\operatorname{Re} s > 23/24$.

1.1 Relation to the source and to the formalization

The argument is that of OpenAI's paper *The Quasi-Riemann Hypothesis* [13], which proves the stronger half-plane $\operatorname{Re} s > 11/12$ for every finite-order Hecke L-function over $K = \mathbb{Q}(\sqrt{-3})$, and hence for every Dirichlet L-function. That paper is in turn an intermediate step towards the half-plane $\operatorname{Re} s > 7/8$ of [22]. We follow [13] closely, and much of the text below is taken from it, but we make three simplifications. The first and third do not affect the exponent; the second weakens it.

1. We treat only ζ. The twisting character of [13] is then trivial, and the Hecke L-function that controls the Möbius sums is the Dedekind zeta function $\zeta_K(s) = \zeta(s)L(s, \chi_{-3})$, so no continuation of Hecke L-functions is needed.

2. We fix the parameter $\vartheta = 1/10$ that measures how far the row range of the basic mean square exceeds the column range. The argument of [13] gives the exponent $11/12 + 5\vartheta/12$ for every $0 < \vartheta \le 1/10$; letting $\vartheta \to 0$ gives $11/12$, while our fixed choice gives $23/24$. Fixing ϑ turns every exponent below into a concrete number.

3. We extract the Möbius sum from the mean square using all rows $u = b^6$, $b \in \mathcal{O}$, rather than the rows $u = p^6$ with p prime. This avoids any prime counting; the price is a harmless totient weight $\varphi(\mathfrak{n})/\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})$ in the Möbius sum, which contributes an Euler product that is holomorphic in $\operatorname{Re} s > 0$.

Theorem 1.1 has been formally verified in Lean 4 with Mathlib, using only the standard axioms. The formal proof is organized like this note:

- every numbered result of Section 3 is a Lean theorem in the directory `QRH/Spine`, and the top-level proof of Theorem 1.1 is the same three-line composition as below;
- the main results of Sections 4–7 are theorems of the formalization, which proves them with somewhat different bookkeeping (Remarks 4.6 and 5.5).

Appendix C lists the correspondence. The formalization was obtained from OpenAI's Lean development of [22], which contains this argument at $\vartheta = 1/10$ as an intermediate result.

1.2 Classical inputs

We use the following standard facts without proof. They are the only places where the argument leaves elementary analysis and the arithmetic of $\mathcal{O} = \mathbb{Z}[\omega]$.

A. *Poisson summation* on the lattice $\mathcal{O} \subset \mathbb{C}$ and the evaluation of Gauss sums of primitive characters [14, (3.12) and Theorem 4.5].

B. *Gauss and Jacobi sums, cubic and sextic reciprocity* in $\mathcal{O}$: the Gauss–Jacobi identity, $J(\chi, \chi) = -p$ for a primary prime p and the cubic character $\chi = (\cdot/p)_3$, and the cubic and quadratic reciprocity laws with their supplements [12, Chapters 8 and 9], [14, §3], [5, (1.5)]. The consequences we need are collected in Lemma 4.1; their derivation is recalled in Appendix A.1.

C. *Kubota's cubic theta function* [17]: it is automorphic for a congruence subgroup of $\mathrm{SL}_2(\mathcal{O})$ with the cubic Kubota character. Its Fourier coefficients at the cusp ∞ are normalized cubic Gauss sums (Patterson [23, Theorem 8.1]), and its expansions at the other cusps are given by [5, §5 and Appendix A], following [23, Table III]. We use the normalization of [5, §5.1].

D. *The quadratic large sieve* over K (Heath-Brown [9] for $\mathbb{Q}$; Goldmakher and Louvel [7, Theorem 1.1] for number fields), recorded in Lemma 6.5.

E. *The factorization* $\zeta_K(s) = \zeta(s)L(s, \chi_{-3})$ of the Dedekind zeta function of K, the meromorphic continuation of ζ (with its only pole at $s = 1$), the fact that $L(s, \chi_{-3})$ is entire, and the nonvanishing of $\zeta(s)$ on $\operatorname{Re} s \geq 1$ (see [14, Chapter 5] and [4]).

In the formalization, Inputs A–D and the factorization in Input E are proved within the development, the cubic theta function included (as the residue of a cubic metaplectic Eisenstein series); the analytic facts about ζ and $L(s, \chi_{-3})$ come from Mathlib. That is where most of the formal proof's length goes; the argument of this note is the remaining layer.

1.3 Organization

- Section 2 gives a high-level outline of the argument.
- Section 3 deduces Theorem 1.1 from a mean-square estimate for a family of twisted Möbius sums (Proposition 3.1).
- Section 4 reduces this estimate to a dual mean square with cubic Gauss-sum coefficients.
- Section 5 states the completed mean-square and transfer estimates, combines them into a recursive inequality, and proves Proposition 3.1.
- Sections 6 and 7 prove the completed estimate and the transfer estimate.
- Appendix A contains the arithmetic identities and the theta calculation, Appendix B the separation of variables in smooth weights, and Appendix C the correspondence with the formal proof.

1.4 Notation

Write $\mathcal{O} = \mathcal{O}_K = \mathbb{Z}[\omega]$, where $\omega = e^{2\pi \mathrm{i}/3}$, and $\lambda = 1 + 2\omega$, so that $\lambda^2 = -3$. For $a \in K$, write $\mathrm{N}_{K/\mathbb{Q}}(a) = a\overline{a} = |a|^2$ for its norm. For a nonzero integral ideal $\mathfrak{a} \subseteq \mathcal{O}$, write $\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{a}) = \#(\mathcal{O}/\mathfrak{a})$. We use $\mathbf{1}_{\mathcal{C}}$ for the indicator of a condition $\mathcal{C}$. For a prime ideal represented by p and a nonzero element or ideal a, write $v_p(a)$ for its exponent in the prime factorization of a. Throughout, S is the set of the two prime ideals (λ) and (2) dividing 6. For an ideal or element a, write $(a, S) = 1$ if no prime ideal in S divides a.

We use standard asymptotic notation, writing $f = O(g)$ or $f \ll g$ when $|f| \leq Cg$, where $g \geq 0$ and $C > 0$ is an absolute constant unless otherwise specified. Subscripts indicate possible dependence of the implied constant: for example, $f \ll_{W,\varepsilon} g$ means $|f| \leq C_{W,\varepsilon} g$, where the constant may depend on W, ε but is uniform in all other varying parameters. For nonnegative f, g, we write $f \asymp g$ when $f \ll g$ and $g \ll f$. A dyadic norm range is an interval $R \leq \mathrm{N}_{K/\mathbb{Q}}(a) < 2R$; a sum over dyadic scales uses $R = 2^j$.

We use $D \geq 1$ as an ambient size parameter; in the main argument, it is the original norm scale introduced in the outline below. Auxiliary scales may vary within ranges bounded by fixed powers of D. We write

$$A \preccurlyeq B \quad \text{if} \quad A \ll_{\varepsilon} D^{\varepsilon} B \qquad \text{for every } \varepsilon > 0,$$

uniformly in the varying parameters over their stated ranges. Implied constants may also depend on the fixed data specified in each statement.

2 Outline of the argument

In this section, we sketch the proof of Theorem 1.1, suppressing various coprimality conditions, local factors, and details of smoothing. The precise statements appear in Sections 3–7.

2.1 Step 1: Reduction to power-saving estimates for Möbius sums

The zeros of $\zeta(s)$ are zeros of the Dedekind zeta function $\zeta_K(s) = \zeta(s)L(s, \chi_{-3})$ of $K = \mathbb{Q}(\sqrt{-3})$, and $1/\zeta_K(s) = \sum_{\mathfrak{n}} \mu(\mathfrak{n})\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})^{-s}$ for $\operatorname{Re} s > 1$, where μ is the Möbius function on the nonzero integral ideals of $\mathcal{O}$. We will deduce Theorem 1.1 from a power-saving estimate for smoothed Möbius sums over ideals. All ideal sums run over nonzero integral ideals prime to 6. For a norm scale $D > 0$, consider

$$A_1(D) = \sum_{\mathfrak{n}} \mu(\mathfrak{n}) W(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})/D),$$

where $W \in C_c^\infty((0,\infty);\mathbb{C})$ is a smooth cutoff function. Thus $A_1(D)$ is a smoothed Möbius sum over ideals of norm comparable to D, with roughly D terms. We seek a power saving over this size: for a fixed $\delta > 0$ and every $\varepsilon > 0$,

$$A_1(D) \ll_{W,\varepsilon} D^{1-\delta+\varepsilon} \text{ for every } W \in C_c^\infty((0,\infty);\mathbb{C}). \tag{2.1}$$

The Mellin transform of $D \mapsto A_1(D)$ is $\widehat{W}(s)/\zeta_K(s)$ up to finitely many Euler factors, so (2.1) continues $1/\zeta_K$ holomorphically to $\operatorname{Re} s > 1 - \delta$. This is the smoothed version of the classical relation between Möbius sums and zero-free regions; for ζ itself, the equivalence between the Riemann Hypothesis and the bound $\sum_{n \le x} \mu(n) \ll_\varepsilon x^{1/2+\varepsilon}$ is due to Littlewood [18]; see also [19, §1]. Section 3 gives the argument in the form we use, in which $\mu(\mathfrak{n})$ carries an additional factor $\varphi(\mathfrak{n})/\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})$ (Step 2 explains why).

2.2 Step 2: Embedding the sum in a family

In order to estimate $A_1(D)$, we embed it into a family of such sums, parametrized by $u \in \mathcal{O}_K$, by introducing a sextic twist. We call an element of $\mathcal{O}$ *primary* if it is congruent to 1 modulo 3. The ring $\mathcal{O} = \mathbb{Z}[\omega]$ is Euclidean, so it has unique factorization and every ideal is principal; for an ideal coprime to 3, multiplying any generator by a unique unit gives $n \equiv 1 \pmod 3$, since the six units represent the six invertible residue classes modulo 3. For a primary element n, we write $\mu(n) = \mu((n))$; divisor sums count each ideal divisor once, using its primary generator. For an ideal $\mathfrak{n}$ prime to 6, use its unique primary generator n and write

$$\chi_{\mathfrak{n}}(u) = \chi_n(u) = (u/n)_6.$$

Here $(u/n)_6$ is the sextic residue symbol, extended by zero when $(u, n) \neq 1$.[1] We also write $(u/n)_2$ and $(u/n)_3$ for the quadratic and cubic residue symbols, defined by the same convention with 6 replaced by 2 and 3, respectively. When $(n, 6) = 1$, they equal $\chi_n(u)^3$ and $\chi_n(u)^2$. Restrictions that keep these symbols defined are understood when omitted in the outline. The family is

$$A_u(D) = \sum_{\mathfrak{n}} \mu(\mathfrak{n})\chi_{\mathfrak{n}}(u) W(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})/D). \tag{2.2}$$

Put $\vartheta = 1/10$ and $H = D^{1+\vartheta} = D^{11/10}$. The crucial estimate, Proposition 3.1, is that

$$\sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(u) \le H} |A_u(D)|^2 \ll_{W,\varepsilon} D^{1+\varepsilon} H = D^{21/10+\varepsilon}. \tag{2.3}$$

[1] For a primary prime $p \nmid 6$ and $p \nmid u$, $(u/p)_6$ is the unique sixth root of unity congruent to $u^{(\mathrm{N}_{K/\mathbb{Q}}(p)-1)/6}$ modulo p; set $(u/p)_6 = 0$ when $p \mid u$. Extend multiplicatively in the denominator to define $\chi_n(u) = (u/n)_6$ for primary n coprime to 6. For the unit ideal, $\chi_1(u) = 1$ for every u, including $u = 0$.

In a mean square, we call the variable in the outer sum the *row* and the variable in the inner sum the *column*. In (2.3) these are u and $\mathfrak{n}$, respectively. Up to the factor of D^{ε}, (2.3) can be thought of as saying that the family $\{A_u(D)\}$ exhibits "square-root cancellation" on average (in the L^2 sense) over u. The argument of [13] works for every fixed $0 < \vartheta \le 1/10$. We take $\vartheta = 1/10$, the value in the formalization, whose fixed-margin descent needs $\vartheta > 1/20$ (Remark 5.5); a smaller ϑ would give a better exponent at the price of a longer descent.

The mean-square estimate (2.3) gives a power saving for the Möbius sum because the rows $u = b^6$ see it almost untwisted: $\chi_n(b^6) = \mathbf{1}_{(b,n)=1}$. Average these rows over $\mathrm{N}_{K/\mathbb{Q}}(b) \le Y$, with $Y = H^{1/6}$, against a smooth bump. Counting the b prime to a given n gives about $Y\varphi(n)/\mathrm{N}_{K/\mathbb{Q}}(n)$ of them, so the average is, with a nonzero constant $\widehat{V}(0)$ (see Section 3),

$$\sum_{b} V\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(b)}{Y}\Big) A_{b^6}(D) = Y\widehat{V}(0)\, A^{\varphi}(D) + O(D^{1+\varepsilon}), \tag{2.4}$$

where

$$A^{\varphi}(D) = \sum_{\mathfrak{n}} \mu(\mathfrak{n}) \frac{\varphi(\mathfrak{n})}{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})} W\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})}{D}\Big).$$

There are about Y rows b^6, all of norm at most $Y^6 = H$, so Cauchy–Schwarz and (2.3) bound the left side by $\sqrt{Y \cdot D^{1+\varepsilon} H}$. Hence

$$A^{\varphi}(D) \ll \frac{(YD^{1+\varepsilon}H)^{1/2}}{Y} + \frac{D^{1+\varepsilon}}{Y} \ll \big(D^{1+\varepsilon}H^{5/6}\big)^{1/2} = D^{23/24+\varepsilon},$$

since the second term, $D^{49/60+\varepsilon}$, is smaller. For general ϑ the same computation gives $D^{11/12+5\vartheta/12+\varepsilon}$; this is where $\vartheta < 1/5$ is needed. The weight $\varphi(\mathfrak{n})/\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})$ changes the Dirichlet series only by an Euler product that converges absolutely in $\mathrm{Re}\, s > 0$, so the saving for A^{φ} serves as well as (2.1). Therefore it remains to establish (2.3).

2.3 Step 3: Poisson summation

We turn to the task of proving the mean-square estimate (2.3). Now that we have introduced the row variable u, we can apply Poisson summation in this variable. This introduces sextic Gauss sums, arising from the Fourier transforms of the sextic characters. Upon application of the Gauss–Jacobi identities, these sextic Gauss sums absorb the factor of μ and turn into cubic Gauss sums. We interpret the resulting cubic Gauss sums as coefficients of Kubota's cubic theta function. In the next step, we use the automorphy of this theta function. For now, we explain the appearance of these Gauss sums in more detail.

Define the additive character

$$e(z) = \exp(4\pi \mathrm{i}\ \mathrm{Im}\, z/\sqrt{3}) \quad (z \in \mathbb{C}).$$

For a squarefree Eisenstein integer $n \equiv 1 \pmod 3$ prime to 6 and an integer j, define the normalized Gauss sums

$$\gamma_j(n) = \frac{1}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(n)}} \sum_{x \bmod n} \chi_n(x)^j e(x/n). \tag{2.5}$$

For $j = -1$, we interpret χ_n^{-1} as the conjugate character $\overline{\chi_n}$.

Consider the classical identity

$$\left(\frac{-1}{m}\right) = \left(\frac{1}{\sqrt{m}} \sum_{x \bmod m} \left(\frac{x}{m}\right) e^{2\pi i x/m}\right)^2,$$

where m is an odd squarefree positive integer and the symbols are Jacobi symbols. We would like a similar decomposition for μ instead of $\left(\frac{-1}{m}\right)$. Following the work of Hasse [8, pp. 443–445] and Heath-Brown [10, (2)], we derive the following identity in Appendix A.1, valid for squarefree primary n away from a fixed set of excluded primes:

$$\mu(n)\gamma_{-1}(n) = \chi_n(-1)G(n)^{-1}\overline{\alpha(n)}\gamma_2(n), \qquad \alpha(n) = \frac{n}{|n|}, \quad G(n) = \overline{\chi_n(4)}\gamma_3(n).$$

Here $G(n)$ is a fixed ray-class factor. The factor $\gamma_{-1}(n)$ comes from Poisson summation; as promised, it combines with $\mu(n)$ to produce the cubic coefficient $\overline{\alpha(n)}\gamma_2(n)$ (up to ray-class factors). Further details are given in Section 4.

We call the sums obtained after applying Poisson summation *dual sums*.[2] Rearranging these dual sums reduces the problem of estimating $\sum_u |A_u(D)|^2$ to estimating the following family of column sums (for clarity, we have suppressed auxiliary twists and coprimality conditions):

$$B_h(X) = \sum_{\substack{n\in\mathcal{O}\\ n\equiv 1\ (3)\\ n\text{ squarefree}}} \overline{\alpha(n)}\gamma_2(n)\xi(n)\chi_n(h)U(\mathrm{N}_{K/\mathbb{Q}}(n)/X). \tag{2.6}$$

Here h is the new row variable, Fourier dual to u, while U is a smooth weight restricting the column variable n to norm comparable to X. Here ξ ranges over a finite set of ray class characters of fixed modulus; these twists absorb the residue-class dependence of the reciprocity and Gauss-sum factors below.[3] For each choice of ξ, this defines a family of sums $B_h(X)$.

After treating the diagonal and separating the smooth weights, we roughly get that

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(u)\le H} |A_u(D)|^2 \preccurlyeq DH + \frac{H}{D}\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |B_h(D)|^2. \tag{2.7}$$

Here $\mathcal{H} \asymp D^2/H$ is the dual norm scale. This schematic comparison suppresses the common factors arising when the square is expanded. Thus the desired estimate is

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |B_h(D)|^2 \preccurlyeq D^2. \tag{2.8}$$

2.4 Step 4: Relation to cubic Gauss sums and Kubota's cubic theta function

To estimate (2.8), we first identify the cubic Gauss coefficients with Fourier coefficients of a theta function. Its transformation law applies to a sum with extra cube factors, which we will remove in Step 5.

Let $\theta(z, v)$ be Kubota's cubic theta function on hyperbolic three-space ($z \in \mathbb{C}$, $v \in \mathbb{R}_{>0}$), obtained as a residue of a cubic metaplectic Eisenstein series [17, 23]. We use the normalization of [5, §5.1, (5.6)–(5.8)], recalled below in (6.1).

Put $\lambda = 1+2\omega$. For primary elements $n, b \in \mathcal{O} = \mathbb{Z}[\omega]$, with n squarefree and $(nb, 6) = 1$, let $c_\theta(nb^3)$ denote the Fourier coefficient of $\overline{\theta}$ indexed by $\lambda^{-3}nb^3$ in its expansion in z. Patterson's formula [23, Theorem 8.1], recorded in [5, (5.7)], gives

$$c_\theta(nb^3) = 3^{5/2}|b|\,\overline{\chi_n(\lambda)^2}\gamma_2(n). \tag{2.9}$$

Taking $b = 1$ in (2.9) and substituting into (2.6) yields

$$B_h(X) = 3^{-5/2}\sum_{\substack{n\in\mathcal{O}\\ n\equiv 1\ (3)\\ n\text{ squarefree}}} c_\theta(n)\overline{\alpha(n)}\chi_n(\lambda)^2\xi(n)\chi_n(h)U(\mathrm{N}_{K/\mathbb{Q}}(n)/X). \tag{2.10}$$

[2] For this terminology, see [25, p. 440], [16, §2.3], and [6, arXiv abstract].

[3] The ray class group modulo an ideal $\mathfrak{m}$ is the group of fractional ideals prime to $\mathfrak{m}$, modulo principal ideals generated by elements congruent to 1 modulo $\mathfrak{m}$. A ray class character is a character of this finite group. For an element prime to $\mathfrak{m}$, its ray class means the class of its principal ideal.

Thus (2.10) expresses $B_h(X)$ as a smoothed, twisted sum of the squarefree-index coefficients of $\overline{\theta}$. This connection was used by Heath-Brown and Patterson to study Kummer sums [11].

To use the summation formula for theta coefficients, we embed the sum (2.10) in a completed sum, by including terms indexed by nb^3. This use of cube completion follows Dunn and Radziwiłł [5, Lemma 5.4 and Proposition 5.3], with the underlying theta coefficients given by Patterson [23, Theorem 8.1]. For fixed h, ξ, and U, define

$$T_h(X) := \frac{1}{3^{5/2}\sqrt{X}} \sum_{\substack{n,b\in\mathcal{O}\\ n,b\equiv 1\ (3)\\ n\ \text{squarefree}}} c_\theta(nb^3)\overline{\alpha(nb^3)}\chi_{nb^3}(\lambda)^2\xi(nb^3)\chi_{nb^3}(h)U(\mathrm{N}_{K/\mathbb{Q}}(nb^3)/X). \tag{2.11}$$

The $b = 1$ terms are exactly $X^{-1/2}B_h(X)$; the other terms supply the cube indices in the theta expansion. The first goal is to bound the mean square of $T_h(X)$ over h.

The automorphy of θ gives a summation formula that transforms the completed column sum $T_h(X)$, with the row h held fixed, into dual sums of theta coefficients with new smooth weights and character twists. A cusp is represented by a boundary point in $K \cup \{\infty\}$. A *cusp expansion* is the Fourier expansion in the horizontal variable after a change of coordinates taking infinity to that point. We call its Fourier coefficients *cusp coefficients*. Our starting point is a variant (established in Appendix A.2) of the theta transformation of Dunn and Radziwiłł [5, §5], extending work of Patterson [23] and Yoshimoto [28]. The key point is how the character twists change. Suppose for illustration that h is squarefree and primary, with $(h, 6) = 1$. Then, by Proposition 6.2, the transformed expression is a finite linear combination of sums of the form

$$\sum_{0\neq m\in\mathcal{O}} \frac{d_\theta(m)\alpha(m)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(m)}}\chi_h(m)^3 V_*^\sharp\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(m)X}{\mathrm{N}_{K/\mathbb{Q}}(h)^2}\Big). \tag{2.12}$$

The dual column index is m, while h remains the row index. Here $d_\theta(m)$ denotes the coefficient at the Fourier index $\lambda^{-4}m$ in a cusp expansion of $\overline{\theta}$, and $V_*^\sharp$ is the transform in (6.6), applied to $V_*(y) = y^{1/2}U(y)$. We have suppressed a finite sum over these cusp expansions, fixed periodic twists, bounded prefactors, and fixed scale constants. Within each fixed ray class of h, the coefficient sequences and transformed weights are independent of h by Lemma 6.3. Proposition 6.2 gives the precise formula, writing $d(\ell)$ for the cusp coefficient at $\ell = \lambda^{-4}m$.

Since $V_*^\sharp$ decays rapidly at infinity, the effective norm range in (2.12) is $\mathrm{N}_{K/\mathbb{Q}}(m) \ll \mathrm{N}_{K/\mathbb{Q}}(h)^2/X$, in place of the original range $\mathrm{N}_{K/\mathbb{Q}}(nb^3) \asymp X$. These dual sums arise from the theta transformation and involve theta coefficients on the transformed scale, now twisted by the quadratic character χ_h^3. This character arises because, at each prime $p \mid h$, the Fourier and theta factors combine as

$$\chi_p^{-1}\chi_p^{-2} = \chi_p^{-3} = \chi_p^3. \tag{2.13}$$

Now a key point is that since χ_p^3 is quadratic, Goldmakher and Louvel's quadratic large sieve [7, Theorem 1.1 and Corollary 1.2] (a generalization of Heath-Brown's quadratic large sieve [9] to number fields) bounds the mean square of the sums in (2.12) as h varies. For squarefree quadratic families with row and column norm ranges M, L, the quadratic large sieve gives the factor $M + L$, up to $(ML)^\varepsilon$. Crucially, this avoids the additional term $(ML)^{2/3}$ in Blomer, Goldmakher, and Louvel's general higher-order large sieve [3, Theorem 1.3].

We then apply Cauchy–Schwarz in the cube variable and the quadratic large sieve in the squarefree column variable to estimate the mean square of $T_h(X)$. The details are given in the proof of Proposition 5.2, which gives, for $\mathcal{H}, X \geq 1$,

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |T_h(X)|^2 \ll_{\varepsilon,\xi,U} (\mathcal{H}X)^\varepsilon\Big(\mathcal{H} + \frac{\mathcal{H}^2}{X}\Big). \tag{2.14}$$

At $X = D$ and $\mathcal{H} \le D$, the heuristic comparison $B_h(D) \approx \sqrt{D}\, T_h(D)$ would therefore give the desired D^2 bound. The remaining task is to justify the corresponding mean-square bound by removing the cube factors.

2.5 Step 5: Removing the cube factors

We now pass from a mean-square bound for $T_h(X)$ to one for $X^{-1/2}B_h(X)$, with both averages taken over h. Note that one cannot simply discard the terms with $b \neq 1$, because the contributions from different b can cancel. We begin by undoing the addition of cube factors using Möbius inversion. Related completions appear in Patterson [23, Theorem 6.1] and Heath-Brown [10, §3], with explicit removal of the cube factors by Möbius inversion in Dunn and Radziwiłł [5, Proposition 5.3 and (8.2)].

Fix h and ξ, and write $P_h(X) = X^{-1/2}B_h(X)$ for the $b = 1$ part of $T_h(X)$. Substituting the coefficient formula (2.9) into (2.11) gives

$$T_h(X) = \sum_{\substack{b \in \mathcal{O} \\ b \equiv 1 \ (3)}} \frac{w_h(b)}{\mathrm{N}_{K/\mathbb{Q}}(b)}\, P_h\Big(\frac{X}{\mathrm{N}_{K/\mathbb{Q}}(b)^3}\Big), \tag{2.15}$$

where

$$w_h(b) = \overline{\alpha(b)}^{\,3}\xi(b)^3\chi_b(h)^3. \tag{2.16}$$

Each factor in (2.16) is completely multiplicative in b. Thus $w_h(bc) = w_h(b)w_h(c)$ even when b, c share prime factors, and $|w_h(b)| \le 1$. Möbius inversion (cf. [14, §1.3]) gives

$$P_h(X) = \sum_{\substack{d \in \mathcal{O} \\ d \equiv 1 \ (3)}} \frac{\mu(d)w_h(d)}{\mathrm{N}_{K/\mathbb{Q}}(d)}\, T_h\Big(\frac{X}{\mathrm{N}_{K/\mathbb{Q}}(d)^3}\Big). \tag{2.17}$$

Indeed, substituting (2.15) into (2.17) and grouping by the total cube index b gives the factor $w_h(b)\sum_{d|b}\mu(d)$, which is 1 for $b = 1$ and 0 otherwise.

The objective is now an estimate for P_h, rather than for the completed sum T_h. For the simplified family and the parameter ranges arising from Step 3, the required bound is

$$\sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(h) \ll \mathcal{H}} |P_h(X)|^2 \ll_{\varepsilon,\xi,U} (\mathcal{H}X)^{\varepsilon}X. \tag{2.18}$$

Since $B_h(X) = \sqrt{X}\, P_h(X)$, this is equivalent to a bound of size X^2, up to the same small power, for the mean square of $B_h(X)$.

For $1 \le H_c \le X^{1/3}$, let $P_{h,\le H_c}(X)$ be the part of (2.17) with $\mathrm{N}_{K/\mathbb{Q}}(d) \le H_c$. Applying weighted Cauchy–Schwarz for each fixed h, then summing over h and using (2.14), gives

$$\begin{aligned}
&\sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(h) \ll \mathcal{H}} |P_{h,\le H_c}(X)|^2 \\
&\quad \le \Big(\sum_{\mathrm{N}_{K/\mathbb{Q}}(e) \le H_c} \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(e)}\Big)\Big(\sum_{\mathrm{N}_{K/\mathbb{Q}}(d) \le H_c} \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(d)} \sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(h) \ll \mathcal{H}} |T_h\big(X/\mathrm{N}_{K/\mathbb{Q}}(d)^3\big)|^2\Big) \\
&\quad \preccurlyeq \sum_{\mathrm{N}_{K/\mathbb{Q}}(d) \le H_c} \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(d)}\Big(\mathcal{H} + \frac{\mathcal{H}^2\mathrm{N}_{K/\mathbb{Q}}(d)^3}{X}\Big) \preccurlyeq \mathcal{H} + \frac{\mathcal{H}^2H_c^3}{X}.
\end{aligned}$$

Thus

$$\sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(h) \ll \mathcal{H}} |P_{h,\le H_c}(X)|^2 \ll_{\varepsilon,\xi,U} (\mathcal{H}X)^{\varepsilon}\Big(\mathcal{H} + \frac{\mathcal{H}^2H_c^3}{X}\Big). \tag{2.19}$$

For $\mathcal{H} \le X$, the estimate (2.19) is within the target (2.18) provided $\mathcal{H}^2 H_c^3/X \le X$. Thus we apply the completed mean-square bound directly only up to the cutoff

$$H_c = \min\Big\{ X^{1/3}, \Big(\frac{X}{\mathcal{H}}\Big)^{2/3} \Big\}. \tag{2.20}$$

At the basic scales from Step 3,

$$X \asymp D, \qquad \mathcal{H} \asymp D^{1-\vartheta} = D^{9/10}, \qquad H_c \asymp D^{2\vartheta/3} = D^{1/15}.$$

The inverse sum can extend to $\mathrm{N}_{K/\mathbb{Q}}(d) \asymp D^{1/3}$, so we must still control the larger divisors.

Write $\tau_{\mathrm{div}}(b)$ for the number of nonzero integral ideal divisors of (b). Let $P_{h,>H_c}(X)$ denote the terms with $\mathrm{N}_{K/\mathbb{Q}}(d) > H_c$ in (2.17), and put $L_b = X/\mathrm{N}_{K/\mathbb{Q}}(b)^3$. Substituting (2.15) into the truncated inversion formula for $P_{h,>H_c}(X)$ obtained from (2.17), and grouping by $b = dc$, gives

$$\begin{aligned} P_{h,>H_c}(X) &= \sum_{\mathrm{N}_{K/\mathbb{Q}}(d)>H_c} \sum_{c} \frac{\mu(d) w_h(d) w_h(c)}{\mathrm{N}_{K/\mathbb{Q}}(d)\mathrm{N}_{K/\mathbb{Q}}(c)} P_h\Big(\frac{X}{\mathrm{N}_{K/\mathbb{Q}}(dc)^3}\Big) \\ &= \sum_{\mathrm{N}_{K/\mathbb{Q}}(b)>H_c} \frac{\beta_0(b)\chi_b(h)^3}{\mathrm{N}_{K/\mathbb{Q}}(b)} P_h(L_b), \end{aligned}$$

where all indices are primary and

$$\beta_0(b) = \overline{\alpha(b)}^{\,3} \xi(b)^3 \sum_{\substack{d \mid b \\ \mathrm{N}_{K/\mathbb{Q}}(d)>H_c}} \mu(d), \qquad |\beta_0(b)| \le \tau_{\mathrm{div}}(b).$$

The support of U restricts $\mathrm{N}_{K/\mathbb{Q}}(b) \ll_U X^{1/3}$. Weighted Cauchy–Schwarz for each h, followed by summation, gives

$$\begin{aligned} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_{h,>H_c}(X)|^2 &\le \Big(\sum_b \frac{\tau_{\mathrm{div}}(b)}{\mathrm{N}_{K/\mathbb{Q}}(b)}\Big) \sum_b \frac{\tau_{\mathrm{div}}(b)}{\mathrm{N}_{K/\mathbb{Q}}(b)} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_h(L_b)|^2 \\ &\preccurlyeq \sup_b \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_h(L_b)|^2. \end{aligned} \tag{2.21}$$

Put $a_\xi(n) = \overline{\alpha(n)}\gamma_2(n)\xi(n)$. Write

$$E(\mathcal{H}, X) := \frac{1}{X} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} \Big| \sideset{}{^*}\sum_n a_\xi(n)\chi_n(h) U(\mathrm{N}_{K/\mathbb{Q}}(n)/X) \Big|^2.$$

A star restricts an index to squarefree primary elements. By (2.6) and the definition of P_h, we have

$$E(\mathcal{H}, X) = \frac{1}{X} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |B_h(X)|^2 = \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(h)\ll\mathcal{H}} |P_h(X)|^2.$$

At the scales from Step 3, our goal is to prove

$$E(\mathcal{H}, X) \preccurlyeq X. \tag{2.22}$$

At $X = D$, this is precisely the bound (2.8) for the mean square of $B_h(D)$. Substitution into the Poisson comparison (2.7) then gives the required original mean-square estimate (2.3). Combining (2.19) for $P_{h,\le H_c}$ (the terms with $\mathrm{N}_{K/\mathbb{Q}}(d) \le H_c$) and (2.21) for $P_{h,>H_c}$ (the terms with $\mathrm{N}_{K/\mathbb{Q}}(d) > H_c$), with the cutoff (2.20), gives

$$E(\mathcal{H}, X) \preccurlyeq X + \sup_{b:\, 1 \le L_b \le X/H_c^3} E(\mathcal{H}, L_b).$$

It therefore remains to prove $E(\mathcal{H}, L_b) \preccurlyeq X$ for $1 < L_b \le X/H_c^3$. Here the row range $\mathcal{H}$ stays fixed, and the required bound is still of size X even though the column scale has decreased to L_b.

We now expand the square and apply Poisson summation in h as before; we now record the proof in terms of $E(\mathcal{H}, L_b)$. The Gauss-sum coefficients become Möbius coefficients, and the new row variable y has norm at most about $L_b^2/\mathcal{H}$. After separating the weights, this gives schematically

$$E(\mathcal{H}, L_b) \preccurlyeq X + \frac{\mathcal{H}}{L_b^2} \sum_{0<\mathrm{N}_{K/\mathbb{Q}}(y)\ll L_b^2/\mathcal{H}} |M(y)|^2,$$

$$M(y) = \sideset{}{^*}\sum_{\mathrm{N}_{K/\mathbb{Q}}(n)\asymp L_b} \mu(n)\xi_1(n)\overline{\chi_n(y)}U_1(\mathrm{N}_{K/\mathbb{Q}}(n)/L_b).$$

Here ξ_1 is another fixed ray class character and U_1 is a smooth compactly supported weight produced by separating the variables. The term X includes the diagonal and zero-frequency contributions, using $\mathcal{H} \le X$.

Since $L_b = X/\mathrm{N}_{K/\mathbb{Q}}(b)^3 \le X$, we may enlarge the nonnegative sum over y to

$$\mathrm{N}_{K/\mathbb{Q}}(y) \ll \mathcal{Y}, \qquad \mathcal{Y} = \frac{XL_b}{\mathcal{H}} \ge \frac{L_b^2}{\mathcal{H}}.$$

The purpose of this enlargement is that a second application of Poisson summation gives a shorter row range:

$$\mathcal{H}' = \frac{L_b^2}{\mathcal{Y}} = \frac{\mathcal{H}L_b}{X}.$$

The coefficients return to cubic Gauss-sum coefficients, and

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(y)\ll\mathcal{Y}} |M(y)|^2 \preccurlyeq YL_b + \mathcal{Y}E'(\mathcal{H}', L_b).$$

Here E' has the same form as E, with possibly different smooth weights and fixed characters; YL_b accounts for the zero-frequency contribution. Consequently,

$$E(\mathcal{H}, L_b) \preccurlyeq X + \frac{X}{L_b}E'\Big(\frac{\mathcal{H}L_b}{X}, L_b\Big), \qquad \frac{E(\mathcal{H}, L_b)}{X} \preccurlyeq 1 + \frac{E'(\mathcal{H}L_b/X, L_b)}{L_b}.$$

Thus it suffices to prove

$$E'\Big(\frac{\mathcal{H}L_b}{X}, L_b\Big) \preccurlyeq L_b.$$

This is the original type of estimate at smaller parameters:

$$(\mathcal{H}', X') = \Big(\frac{\mathcal{H}}{\mathrm{N}_{K/\mathbb{Q}}(b)^3}, \frac{X}{\mathrm{N}_{K/\mathbb{Q}}(b)^3}\Big), \qquad \frac{\mathcal{H}'}{X'} = \frac{\mathcal{H}}{X}.$$

Both scales decrease while their ratio stays fixed. The factor X/L_b in the preceding inequality is exactly what converts the new target bound L_b into the required bound X. These two Poisson summations give the *transfer estimate*: a bound for the remaining mean square in terms of new mean squares of the same type. For related uses of an enlarged summation range, see Goldmakher–Louvel [7, Lemma 4.4 and the proof of Theorem 4.1], following Heath-Brown [9, Lemma 9].

Combining the bounds (2.19) and (2.21) from the first part of Step 5 with the transfer estimate above gives the recursive bound

$$\frac{E(\mathcal{H}, X)}{X} \preccurlyeq 1 + \sup_{\substack{\mathrm{N}_{K/\mathbb{Q}}(b)>H_c \\ L_b>1}} \frac{E'(\mathcal{H}', X')}{X'}, \qquad X' = L_b, \quad \mathcal{H}' = \frac{\mathcal{H}L_b}{X}.$$

For each b in this supremum, the cutoff (2.20) gives $\mathcal{H}' < \mathcal{H}(\mathcal{H}/X)^2 \ll D^{-2\vartheta}\mathcal{H} = D^{-1/5}\mathcal{H}$. Since $\mathcal{H}'/X' = \mathcal{H}/X$, the same contraction applies at every step. After a bounded number of steps, every resulting mean square either has an empty remainder or has row parameter at most 1, where counting gives the desired bound at its reduced scales. Applying the recursive inequality back through these steps proves (2.22) for the original $E(\mathcal{H}, X)$, completing the sketch of the proof.

The preceding sketch suppresses auxiliary twists and common factors for the purpose of illustration. To carry out this argument with the auxiliary twists included, we use the family

$$\mathcal{E}(\mathcal{H}, X, F) = \frac{1}{XF} \sideset{}{^*}\sum_{\mathrm{N}_{K/\mathbb{Q}}(f) \asymp F} \sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(k) \ll \mathcal{H}} \Big| \sideset{}{^*}\sum_{n} a_\xi(n) \chi_n(k) \chi_n(f)^4 U(\mathrm{N}_{K/\mathbb{Q}}(n)/X) \Big|^2$$

and prove $\mathcal{E}(\mathcal{H}, X, F) \preccurlyeq XF$ in the parameter ranges of Proposition 5.1. Unlike the sketch above, the full argument must also handle the common factors and the resulting dyadic ranges. The precise family is defined in (4.9), and Proposition 5.4 states the transfer estimate. Combining it with the bounds for the two parts of the inverse sum in Step 5 reduces the row range at each step. Section 5 proves the desired bound by a finite iteration, following the admissible-exponent method of Heath-Brown [9, Lemma 8 and §8], [10, Lemma 9]; see also [7, Theorem 4.1] and [3, §3.2].

3 From the mean-square estimate to the zero-free region

We carry out Steps 1 and 2 of the outline: we extract cancellation in a Möbius sum from a mean-square estimate for the sextic family, and then use a Mellin transform to deduce nonvanishing. Recall that $S = \{(\lambda), (2)\}$ is the set of prime ideals dividing 6. Throughout this section, $\mathfrak{n}$ runs over the nonzero ideals of $\mathcal{O}$ with $(\mathfrak{n}, S) = 1$, represented by their primary generators n, and $W \in C_c^\infty((0,\infty); \mathbb{C})$. Define

$$A_u(D) := \sum_{(\mathfrak{n},S)=1} \mu(\mathfrak{n}) \chi_{\mathfrak{n}}(u) W(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})/D) \qquad (u \in \mathcal{O}), \tag{3.1}$$

$$A^\varphi(D) := \sum_{(\mathfrak{n},S)=1} \mu(\mathfrak{n}) \frac{\varphi(\mathfrak{n})}{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})} W(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})/D), \qquad \frac{\varphi(\mathfrak{n})}{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})} = \prod_{\mathfrak{p} \mid \mathfrak{n}} \Big(1 - \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})}\Big). \tag{3.2}$$

The key estimate is the following square-root cancellation on average over the rows u.

Proposition 3.1. *Let $W \in C_c^\infty((0,\infty); \mathbb{C})$ and $\varepsilon > 0$. For $D \geq 1$,*

$$\sum_{\substack{u \in \mathcal{O} \\ 0 < \mathrm{N}_{K/\mathbb{Q}}(u) \leq D^{11/10}}} |A_u(D)|^2 \ll_{W,\varepsilon} D^{21/10+\varepsilon}. \tag{3.3}$$

Here $D^{11/10} = D^{1+\vartheta}$ with $\vartheta = 1/10$, and the right side is $D^{1+\varepsilon} \cdot D^{1+\vartheta}$: each of the roughly $D^{1+\vartheta}$ rows contributes about D, the size of a sum of D unimodular terms with random signs. The proof of Proposition 3.1 occupies Sections 4–7 and is completed in Section 5.6. We first deduce Theorem 1.1 from it.

3.1 Extraction

The rows $u = b^6$, $b \in \mathcal{O}$, see the Möbius sum almost untwisted. Indeed, for a prime $\mathfrak{p} \notin S$ the character $\chi_{\mathfrak{p}}$ has order six, so

$$\chi_{\mathfrak{n}}(b^6) = \mathbf{1}_{(b,\mathfrak{n})=1} \qquad (b \in \mathcal{O}). \tag{3.4}$$

Averaging over b therefore replaces $\chi_{\mathfrak{n}}(u)$ by the density of the b prime to $\mathfrak{n}$, which is $\varphi(\mathfrak{n})/\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})$.

Proposition 3.2. *Let $W \in C_c^\infty((0,\infty);\mathbb{C})$ and $\varepsilon > 0$, and suppose that (3.3) holds for W with 2ε in place of ε. Then, for $D \geq 1$,*

$$A^\varphi(D) \ll_{W,\varepsilon} D^{23/24+\varepsilon}.$$

Proof. Fix a Schwartz function $V : \mathbb{R} \to \mathbb{C}$ supported in $[1/2, 1]$, with $|V| \leq 1$ and

$$\widehat{V}(0) := \frac{2}{\sqrt{3}} \int_{\mathbb{C}} V(|z|^2)\, dx\, dy \neq 0,$$

the value at 0 of the radial Fourier transform of Lemma 4.2 (with the measure for which $\mathcal{O}$ has covolume one). Put $Y = D^{11/60}$, so that $Y^6 = D^{11/10}$, and consider

$$\mathcal{V}(D) := \sum_{b \in \mathcal{O}} V\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(b)}{Y}\Big) A_{b^6}(D).$$

Evaluation. By (3.4),

$$\mathcal{V}(D) = \sum_{(\mathfrak{n},S)=1} \mu(\mathfrak{n}) W(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})/D) \sum_{\substack{b \in \mathcal{O} \\ (b,\mathfrak{n})=1}} V\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(b)}{Y}\Big).$$

For squarefree $\mathfrak{n}$, inclusion–exclusion over $\mathfrak{d} \mid \mathfrak{n}$ writes the inner sum as

$$\sum_{\mathfrak{d} \mid \mathfrak{n}} \mu(\mathfrak{d}) \sum_{c \in \mathcal{O}} V\big(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{d})\mathrm{N}_{K/\mathbb{Q}}(c)/Y\big).$$

By Poisson summation (Input A), the sum over c is $(Y/\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{d}))\widehat{V}(0) + O_V(1)$: the zero frequency gives the main term, the other frequencies are $O_V(1)$ by the rapid decay of $\widehat{V}$, and when $\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{d}) > Y$ only $c = 0$ is in the support, where $V(0) = 0$. Summing over $\mathfrak{d}$,

$$\sum_{\substack{b \in \mathcal{O} \\ (b,\mathfrak{n})=1}} V\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(b)}{Y}\Big) = Y\widehat{V}(0)\frac{\varphi(\mathfrak{n})}{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})} + O_V\big(2^{\omega(\mathfrak{n})}\big).$$

Since $2^{\omega(\mathfrak{n})} \ll_\varepsilon \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})^\varepsilon$ and $W(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})/D)$ restricts to $\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n}) \ll_W D$, this gives

$$\mathcal{V}(D) = Y\widehat{V}(0)A^\varphi(D) + O_{W,V,\varepsilon}(D^{1+\varepsilon}). \tag{3.5}$$

Cauchy–Schwarz. The terms of $\mathcal{V}(D)$ have $Y/2 \leq \mathrm{N}_{K/\mathbb{Q}}(b) \leq Y$, so $b \neq 0$. There are $O(Y)$ such b, the map $b \mapsto b^6$ is at most six-to-one, and its values $u = b^6$ satisfy $0 < \mathrm{N}_{K/\mathbb{Q}}(u) \leq Y^6 = D^{11/10}$. Hence, by (3.3) with 2ε,

$$|\mathcal{V}(D)|^2 \leq \Big(\sum_{\mathrm{N}_{K/\mathbb{Q}}(b) \leq Y} 1\Big) \sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(b) \leq Y} |A_{b^6}(D)|^2 \ll Y \sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(u) \leq D^{11/10}} |A_u(D)|^2 \ll_{W,\varepsilon} Y D^{21/10+2\varepsilon}.$$

Conclusion. Since $Y \cdot D^{21/10} = D^{137/60}$, combining with (3.5) gives

$$Y|\widehat{V}(0)|\,|A^\varphi(D)| \ll D^{137/120+\varepsilon} + D^{1+\varepsilon} \ll D^{137/120+\varepsilon},$$

and dividing by $Y = D^{22/120}$ leaves $A^\varphi(D) \ll D^{115/120+\varepsilon} = D^{23/24+\varepsilon}$. □

3.2 The Mellin transform

For $\operatorname{Re} s > 0$ put

$$E_\varphi(s) := \prod_{\mathfrak{p}\notin S} \frac{1 - \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-s}}{1 - \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-s} + \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-1-s}}, \qquad P_S(s) := \prod_{\mathfrak{p}\in S}\bigl(1 - \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-s}\bigr).$$

Every prime $\mathfrak{p} \notin S$ has $\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p}) \geq 7$, so for $\sigma = \operatorname{Re} s > 0$ the denominator satisfies $|1 - (1 - \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-1})\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-s}| \geq 1 - 7^{-\sigma} > 0$, and each factor of E_φ is $1 + O_\sigma(\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-1-\sigma})$ uniformly in $\mathfrak{p}$. So the product converges absolutely, and E_φ is holomorphic on $\operatorname{Re} s > 0$. The polynomial P_S is entire. For $\operatorname{Re} s > 1$, comparing Euler products gives

$$\sum_{(\mathfrak{n},S)=1} \mu(\mathfrak{n})\frac{\varphi(\mathfrak{n})}{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})}\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})^{-s} = \prod_{\mathfrak{p}\notin S}\Bigl(1 - \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-s} + \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})^{-1-s}\Bigr) = \frac{1}{\zeta_K(s)P_S(s)E_\varphi(s)}, \tag{3.6}$$

where $\zeta_K(s) = \zeta(s)L(s,\chi_{-3})$ (Input E).

Proposition 3.3. *Suppose that $A^\varphi(D) \ll_{W,\varepsilon} D^{23/24+\varepsilon}$ for every $\varepsilon > 0$ and every $W \in C^\infty(\mathbb{R})$ supported in $[1,2]$ (in the formalization: $W(x) = 0$ for $x \leq 1$ and for $x \geq 2$, an equivalent condition for continuous W). Then $\zeta(s) \neq 0$ for $\operatorname{Re} s > 23/24$.*

Proof. For $\operatorname{Re} s \geq 1$ this is classical (Input E). Suppose that $\zeta(\varrho) = 0$ with $23/24 < \operatorname{Re}\varrho < 1$. Choose $0 \neq \phi \in C_c^\infty((1,2))$ with $\phi \geq 0$, and put $W(y) = y^{-\varrho}\phi(y)$. With the Mellin convention $\widehat{W}(s) = \int_0^\infty W(y)y^s\,dy/y$, we have $\widehat{W}(\varrho) = \int \phi(y)\,dy/y > 0$.

The function

$$\mathcal{M}^\varphi(s) = \int_0^\infty A^\varphi(D)D^{-s}\frac{dD}{D}$$

is holomorphic on $\operatorname{Re} s > 23/24$. The hypothesis controls the integral at infinity, and the support of W in $[1,2]$ makes $A^\varphi(D)$ vanish for $D < 1/2$, since every $\mathfrak{n}$ has $\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n}) \geq 1 > 2D$. For $\operatorname{Re} s > 2$ the sum and the integral converge absolutely, and termwise integration followed by (3.6) gives

$$\zeta(s)L(s,\chi_{-3})\,P_S(s)E_\varphi(s)\,\mathcal{M}^\varphi(s) = \widehat{W}(s).$$

Both sides are holomorphic on the connected set $\{\operatorname{Re} s > 23/24\} \setminus \{1\}$. Here $L(s,\chi_{-3})$ is entire and ζ has its only pole at $s = 1$. By the identity theorem the equation holds on that set. At $s = \varrho$ the left side vanishes and the right side does not, a contradiction. □

Proof of Theorem 1.1. Let W be supported in $[1,2]$ and $\varepsilon > 0$. Proposition 3.1, applied with 2ε, and Proposition 3.2 give $A^\varphi(D) \ll_{W,\varepsilon} D^{23/24+\varepsilon}$. Proposition 3.3 concludes. □

4 Poisson summation and the dual mean square

We now turn to the mean-square estimate in Proposition 3.1. Following Step 3 of the outline, we expand the square and apply Poisson summation in the row variable u. The finite Fourier transforms of the characters supply Gauss sums. Arithmetic identities then combine these Gauss sums with the original Möbius coefficients to give the normalized cubic Gauss-sum coefficients that appear in the theta function.

The column indices produced by expanding the square need not be coprime. We first extract their common factor, then use Möbius inversion to separate the remaining coprimality condition. These operations introduce an auxiliary twisting index and an exclusion ideal. The comparison below removes the exclusion without changing the row range or the product of the column and auxiliary scales.

We record the arithmetic and Poisson identities first, then define this family. Proposition 4.5 states the estimate for it that suffices to prove (3.3); the rest of the section proves that implication.

4.1 The arithmetic identities

We first record the identities that convert between Möbius coefficients and normalized cubic Gauss sums. Recall that, for a squarefree primary element n with $(n, S) = 1$,

$$\alpha(n) = \frac{n}{|n|}, \qquad \gamma_j(n) = \frac{1}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(n)}} \sum_{x \bmod n} \chi_n(x)^j e(x/n) \quad (j \in \mathbb{Z}),$$

where $e(z) = \exp(4\pi\mathrm{i}\,\mathrm{Im}\, z/\sqrt{3})$, as in (2.5). Put

$$a_\xi(n) = \overline{\alpha(n)}\gamma_2(n)\xi(n) \qquad (n \text{ squarefree}),$$

where ξ is a fixed ray class character. The character ξ accounts for the residue-class factors in the identities below.

By character orthogonality [14, §3.1], we can absorb functions on a fixed ray class group into a finite sum of twists ξ. Choose a fixed modulus, supported on S, divisible by 12 and sufficiently large that all reciprocity factors below depend only on the corresponding ray classes. We then expand these factors in characters of this finite group.

The following lemma collects standard consequences of the Gauss–Jacobi identities, quadratic Gauss-sum evaluations, and reciprocity.

Lemma 4.1. *On a fixed ray class group, there are a function G with values in $\{z \in \mathbb{C} : |z| = 1\}$ and a symmetric $\{\pm 1\}$-valued bicharacter $\mathcal{R}$. This means that $\mathcal{R}(a, b) = \mathcal{R}(b, a)$ and $\mathcal{R}$ is multiplicative in each argument separately:*

$$\mathcal{R}(aa', b) = \mathcal{R}(a, b)\mathcal{R}(a', b),$$
$$\mathcal{R}(a, bb') = \mathcal{R}(a, b)\mathcal{R}(a, b'),$$

for all classes a, a', b, b' in the ray class group. These functions have the following properties. Let a, b, n be primary elements prime to S, with $(a, b) = 1$ and n squarefree. Then

$$\chi_b(a) = \mathcal{R}(a, b)\chi_a(b), \qquad G(ab) = G(a)G(b)\mathcal{R}(a, b), \tag{4.1}$$
$$\gamma_2(n)^3 = \mu(n)\alpha(n), \qquad \gamma_1(n)\gamma_2(n) = \mu(n)\alpha(n)G(n), \tag{4.2}$$
$$G(n) = \overline{\chi_n(4)}\gamma_3(n), \qquad \gamma_1(n)\gamma_{-1}(n) = \chi_n(-1). \tag{4.3}$$

Consequently

$$\overline{\alpha(n)}\gamma_2(n)\gamma_1(n) = \mu(n)G(n), \tag{4.4}$$
$$\mu(n)\gamma_{-1}(n) = \chi_n(-1)G(n)^{-1}\overline{\alpha(n)}\gamma_2(n), \tag{4.5}$$
$$a_\xi(ab) = a_\xi(a)a_\xi(b)\chi_b(a)^4, \tag{4.6}$$
$$\chi_a(-1)\overline{G(a)}G(b)\mathcal{R}(a, b) = G(ba^{-1}). \tag{4.7}$$

In (4.6), a, b are also squarefree. The identity for $G(ab)$ in (4.1) extends to all classes of the fixed ray class group; the quotient in (4.7) is taken in that group.

The proof, including the dependence of G and $\mathcal{R}$ on fixed ray classes, is given in Appendix A.1. Under Poisson summation, (4.5) converts the Möbius coefficients to $a_\xi(n)$, up to fixed ray class factors, while (4.4) converts them back.

4.2 Poisson summation with excluded primes

The row sum to which we apply Poisson summation will have an additional coprimality restriction. We record the formula with that restriction included, so that its effect on the Fourier frequencies and the normalization is explicit.

For a nonzero ideal $\mathfrak{r}$, write $\operatorname{rad}\mathfrak{r}$ for the product of its distinct prime divisors. We use the additive character e introduced in Step 3. In quotients and Gauss sums, use a fixed generator for each ideal, chosen primary when the ideal is prime to 3.

Lemma 4.2. *Let $\mathcal{H} > 0$, let $\mathfrak{r}$ be a nonzero ideal, and let $\Phi(\mathrm{N}_{K/\mathbb{Q}}(k)/\mathcal{H})$ be a smooth radial Schwartz weight on the row lattice. Let χ be a primitive multiplicative character of $(\mathcal{O}/\mathfrak{m})^\times$, viewed as a function on $\mathcal{O}$ by reduction modulo $\mathfrak{m}$ and extension by zero on nonunits. Write*

$$\gamma(\chi) = \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{m})^{-1/2} \sum_{x \bmod \mathfrak{m}} \chi(x) e(x/\mathfrak{m})$$

for its normalized Gauss sum. Since χ is primitive, its modulus $\mathfrak{m}$ is determined by χ and is suppressed in the notation. Then

$$\begin{aligned} &\sum_k \chi(k) \mathbf{1}_{(k,\mathfrak{r})=1} \Phi(\mathrm{N}_{K/\mathbb{Q}}(k)/\mathcal{H}) \\ &\quad = \frac{\mathcal{H}\gamma(\chi)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{m})}} \sum_{d \mid \mathrm{rad}\,\mathfrak{r}} \frac{\mu(d)\chi(d)}{\mathrm{N}_{K/\mathbb{Q}}(d)} \sum_h \overline{\chi(h)} \widehat{\Phi}\Big(\frac{\mathcal{H}\mathrm{N}_{K/\mathbb{Q}}(h)}{\mathrm{N}_{K/\mathbb{Q}}(d)\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{m})} \Big). \end{aligned} \tag{4.8}$$

Here $\widehat{\Phi}$ is defined by

$$\widehat{\Phi}(|w|^2) = \frac{2}{\sqrt{3}} \int_{\mathbb{C}} \Phi(|z|^2) e(-zw)\, dx\, dy, \qquad z = x + \mathrm{i}y.$$

The measure is normalized so that $\mathcal{O}$ has covolume one. Here $k, h \in \mathcal{O}$, and h is the Fourier frequency. We have

$$\chi \text{ nonprincipal} \quad \Longrightarrow \quad \overline{\chi(0)} = 0.$$

For the principal primitive character ($\chi = \mathbf{1}$, $\mathfrak{m} = 1$), the zero-frequency contribution is

$$\mathcal{H}\widehat{\Phi}(0) \prod_{\mathfrak{p} \mid \mathfrak{r}} \Big(1 - \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{p})}\Big).$$

Proof. Inclusion–exclusion followed by $k = d\ell$ gives

$$\sum_k \chi(k) \mathbf{1}_{(k,\mathfrak{r})=1} \Phi(\mathrm{N}_{K/\mathbb{Q}}(k)/\mathcal{H}) = \sum_{d \mid \mathrm{rad}\,\mathfrak{r}} \mu(d)\chi(d) \sum_\ell \chi(\ell) \Phi\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(\ell)}{\mathcal{H}/\mathrm{N}_{K/\mathbb{Q}}(d)} \Big).$$

Apply lattice Poisson summation [14, Theorem 4.5] in ℓ on residue classes modulo $\mathfrak{m}$, at scale $\mathcal{H}/\mathrm{N}_{K/\mathbb{Q}}(d)$. The primitive Gauss-sum identity [14, (3.12)] evaluates the finite Fourier transform as $\sqrt{\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{m})}\gamma(\chi)\overline{\chi(h)}$ for every h, giving (4.8). For a nonprincipal primitive character, $\overline{\chi(0)} = 0$, so the zero-frequency term vanishes. For the principal character, sum $\mu(d)/\mathrm{N}_{K/\mathbb{Q}}(d)$ over $d \mid \mathrm{rad}\,\mathfrak{r}$ to obtain the displayed zero-frequency contribution. □

4.3 The dual mean squares

We use the following mean square of the column sums with coefficients $a_\xi(n)$.

Definition 4.3. The parameters $\mathcal{H}$, X, and F are the norm scales of k, n, and f, respectively. Here k ranges over elements of $\mathcal{O}$, and a star restricts a sum to squarefree ideals prime to S, represented by their primary generators. For a smooth compactly supported weight W on $(0, \infty)$, define the *dual mean square* by

$$\begin{aligned} &\mathcal{E}(\mathcal{H}, X, F; \xi, W) \\ &\quad = \frac{1}{XF} \sideset{}{^*}\sum_{\substack{F \le \mathrm{N}_{K/\mathbb{Q}}(f) < 2F \\ (f,S)=1}} \sum_{0 < \mathrm{N}_{K/\mathbb{Q}}(k) \le \mathcal{H}} \Big| \sideset{}{^*}\sum_{(n,S)=1} a_\xi(n) \chi_n(k) \chi_n(f)^4 W(\mathrm{N}_{K/\mathbb{Q}}(n)/X) \Big|^2. \end{aligned} \tag{4.9}$$

Below, ξ ranges over all characters of the fixed ray class group chosen above, through which G and $\mathcal{R}$ factor.

Write $\mathcal{E}_{\mathfrak{r}}$ for the same expression with the additional restriction $(n,\mathfrak{r})=1$. This notation is only needed in the Poisson reductions; the following comparison returns to $\mathcal{E}$.

Lemma 4.4. *Let r_0 be the product of the primes dividing $\mathfrak{r}$ outside S. For $\mathcal{H}, X > 0$, $F \geq 1$, and smooth compactly supported W,*

$$\mathcal{E}_{\mathfrak{r}}(\mathcal{H},X,F;\xi,W) \leq \tau_{\mathrm{div}}(r_0)\sum_{d\mid r_0}\mathcal{E}\Big(\mathcal{H},\frac{X}{\mathrm{N}_{K/\mathbb{Q}}(d)},F\mathrm{N}_{K/\mathbb{Q}}(d);\xi,W\Big). \tag{4.10}$$

Proof. Let $C_{\mathfrak{r}}(X;k,f)$ denote the inner column sum defining $\mathcal{E}_{\mathfrak{r}}$, with ξ, W fixed. Inclusion–exclusion, $n=dm$, and (4.6) give

$$\begin{aligned} C_{\mathfrak{r}}(X;k,f) &= \sum_{d\mid r_0}\mu(d)\sideset{}{^*}\sum_{\substack{(n,S)=1\\ d\mid n}} a_\xi(n)\chi_n(k)\chi_n(f)^4 W(\mathrm{N}_{K/\mathbb{Q}}(n)/X) \\ &= \sum_{\substack{d\mid r_0\\(d,f)=1}}\mu(d)a_\xi(d)\chi_d(k)\chi_d(f)^4 C_1(X/\mathrm{N}_{K/\mathbb{Q}}(d);k,df). \end{aligned}$$

Indeed, $\chi_m(d)^4$ enforces $(m,d)=1$ and combines with $\chi_m(f)^4$; terms with $(d,f)\neq 1$ vanish. Each exterior coefficient has modulus at most one. Apply Cauchy–Schwarz in d, then enlarge the injective image $f\mapsto df$ to the squarefree range $F\mathrm{N}_{K/\mathbb{Q}}(d) \leq \mathrm{N}_{K/\mathbb{Q}}(df) < 2F\mathrm{N}_{K/\mathbb{Q}}(d)$. The normalizing product is unchanged: $(X/\mathrm{N}_{K/\mathbb{Q}}(d))(F\mathrm{N}_{K/\mathbb{Q}}(d)) = XF$. □

4.4 Reduction to the dual mean square

The following proposition gives the dual estimate sufficient for (3.3). Its proof applies Poisson summation directly to the original Möbius sums.

Proposition 4.5. *Let $\vartheta = 1/10$ and put $H = D^{1+\vartheta} = D^{11/10}$. For each fixed $C \geq 1$ and all real $B, F \geq 1$, consider the ranges*

$$X = \frac{D}{BF}, \qquad 0 < \mathcal{H} \leq \frac{CD^2}{HB^2}, \qquad XF = \frac{D}{B}. \tag{4.11}$$

Suppose that for every $\varepsilon > 0$ there exists an integer $J = J(\varepsilon) \geq 1$ such that, for every compact interval $I \subset (0,\infty)$ and every smooth W supported in I,

$$\mathcal{E}(\mathcal{H},X,F;\xi,W) \ll_{I,C,\varepsilon} \|W\|^2_{C^J(I)} D^{\varepsilon} XF, \tag{4.12}$$

where

$$\|W\|_{C^J(I)} = \max_{0\leq j\leq J}\sup_{x\in I}|W^{(j)}(x)|.$$

For functions of several variables, the norm uses all partial derivatives of total order at most J. Then the original mean-square estimate in Proposition 3.1 holds.

Proof. Write $I=[a_0,b_0]$. Choose a nonnegative radial Schwartz weight Φ such that $\Phi(x)\geq 1$ on $[0,1]$ and $\operatorname{supp}\widehat{\Phi}\subset[0,C_\Phi]$ for some fixed $C_\Phi>0$. It suffices to prove $\mathcal{M}_D \ll_{I,\varepsilon} \|W\|^2_{C^{J'}(I)} HD^{\varepsilon}$ for some $J'=J'(\varepsilon)$, where

$$\mathcal{M}_D := \frac{1}{D}\sum_{u\in\mathcal{O}}\Phi(\mathrm{N}_{K/\mathbb{Q}}(u)/H)\Big|\sum_{(n,S)=1}\mu(n)\chi_n(u)W(\mathrm{N}_{K/\mathbb{Q}}(n)/D)\Big|^2. \tag{4.13}$$

Conjugate the inner sum, expand the square, and put $g=(n_1,n_2)$ and $n_j=gz_j$. The squarefree indices satisfy $(z_1,z_2)=(z_1z_2,g)=1$, and

$$\overline{\chi_{n_1}(u)}\chi_{n_2}(u)=\mathbf{1}_{(u,g)=1}\overline{\chi_{z_1}(u)}\chi_{z_2}(u).$$

Apply Lemma 4.2 with $\chi=\overline{\chi_{z_1}}\chi_{z_2}$ and exclusion g:

$$\begin{aligned}&\sum_u \Phi(\mathrm{N}_{K/\mathbb{Q}}(u)/H)\mathbf{1}_{(u,g)=1}\chi(u)\\&\quad=\sum_{e\mid g}\frac{H\mu(e)\chi(e)\gamma(\chi)}{\mathrm{N}_{K/\mathbb{Q}}(e)\sqrt{\mathrm{N}_{K/\mathbb{Q}}(z_1)\mathrm{N}_{K/\mathbb{Q}}(z_2)}}\sum_h\overline{\chi(h)}\widehat{\Phi}\Big(\frac{H\mathrm{N}_{K/\mathbb{Q}}(h)}{\mathrm{N}_{K/\mathbb{Q}}(e)\mathrm{N}_{K/\mathbb{Q}}(z_1)\mathrm{N}_{K/\mathbb{Q}}(z_2)}\Big).\end{aligned}\tag{4.14}$$

The zero frequency occurs only when $z_1=z_2=1$ and contributes

$$Z=\frac{H}{D}\widehat{\Phi}(0)\sideset{}{^*}\sum_{(g,S)=1}|W(\mathrm{N}_{K/\mathbb{Q}}(g)/D)|^2\prod_{p\mid g}\Big(1-\frac{1}{\mathrm{N}_{K/\mathbb{Q}}(p)}\Big)\ll_{\Phi,I}H\|W\|_\infty^2.$$

Retain all $h\neq0$, including those with $z_1=z_2=1$.

Expand the fixed ray class function $G(t^{-1})=\sum_\xi c_\xi\xi(t)$. The Chinese remainder theorem and (4.5)–(4.7) give

$$\mu(z_1)\mu(z_2)\gamma(\overline{\chi_{z_1}}\chi_{z_2})=\sum_\xi c_\xi a_\xi(z_1)\overline{a_\xi(z_2)}.\tag{4.15}$$

Write $N=\mathrm{N}_{K/\mathbb{Q}}$ for the remainder of this proof, and put $W_0(x)=x^{-1/2}\overline{W(x)}$. Substituting (4.15) into (4.14), including the normalization $1/D$ in (4.13), gives

$$\mathcal{M}_D-Z=\sum_\xi c_\xi\mathcal{S}_\xi,\qquad|\mathcal{M}_D-Z|\ll\max_\xi|\mathcal{S}_\xi|,$$

since the character sum is fixed and finite. Fix ξ. Using $\overline{\chi_z(e)}=\chi_z(e^5)$, its contribution is

$$\begin{aligned}\mathcal{S}_\xi=\frac{H}{D^2}\sideset{}{^*}\sum_g\sum_{e\mid g}\frac{\mu(e)N(g)}{N(e)}\sum_{h\neq0}\sideset{}{^*}\sum_{\substack{z_1,z_2\\(z_1,z_2)=1\\(z_1z_2,g)=1}}&a_\xi(z_1)\overline{a_\xi(z_2)}\chi_{z_1}(he^5)\overline{\chi_{z_2}(he^5)}\\&\times W_0\Big(\frac{N(gz_1)}{D}\Big)\overline{W_0\Big(\frac{N(gz_2)}{D}\Big)}\widehat{\Phi}\Big(\frac{HN(h)}{N(e)N(z_1)N(z_2)}\Big).\end{aligned}$$

Here and below every starred variable is squarefree, primary, and prime to S, while h,k range over nonzero elements of $\mathcal{O}$. We must show $|\mathcal{S}_\xi|\ll_{I,\varepsilon}\|W\|^2_{C^{J'}(I)}HD^\varepsilon$.

Insert the coprimality identity and change variables:

$$\mathbf{1}_{(z_1,z_2)=1}=\sum_{v\mid z_1,\ v\mid z_2}\mu(v),\qquad z_j=vm_j.$$

By (4.6), the common factor from the two columns satisfies

$$a_\xi(vm)=a_\xi(v)a_\xi(m)\chi_m(v)^4,\qquad|a_\xi(v)\chi_v(he^5)|^2=\mathbf{1}_{(v,h)=1},$$

where $(v,e)=1$. Thus the same sum becomes

$$\begin{aligned}\mathcal{S}_\xi=\frac{H}{D^2}\sideset{}{^*}\sum_{\substack{g,v\\(g,v)=1}}\sum_{e\mid g}\frac{\mu(e)\mu(v)N(g)}{N(e)}\sum_{\substack{h\neq0\\(h,v)=1}}\sideset{}{^*}\sum_{\substack{m_1,m_2\\(m_1m_2,gv)=1}}&a_\xi(m_1)\overline{a_\xi(m_2)}\\&\times\chi_{m_1}(he^5v^4)\overline{\chi_{m_2}(he^5v^4)}W_0\Big(\frac{N(gvm_1)}{D}\Big)\overline{W_0\Big(\frac{N(gvm_2)}{D}\Big)}\\&\times\widehat{\Phi}\Big(\frac{HN(h)}{N(e)N(v)^2N(m_1)N(m_2)}\Big).\end{aligned}$$

There is now no restriction $(m_1,m_2)=1$. Make the bijective change of variables

$$b=g/e,\qquad f=ev,\qquad k=eh,$$
$$e=(f,k),\qquad v=f/e,\qquad g=be,\qquad h=k/e.$$

The resulting b,f are coprime and squarefree, $k\neq 0$ is arbitrary, and

$$gv=bf,\qquad he^5v^4=kf^4,\qquad \mu(e)\mu(v)=\mu(f),\qquad \frac{N(g)}{N(e)}=N(b).$$

Consequently,

$$\begin{aligned}\mathcal{S}_\xi=\frac{H}{D^2}\sideset{}{^*}\sum_{\substack{b,f\\(b,f)=1}}\mu(f)N(b)\sum_{k\neq0}\sideset{}{^*}\sum_{\substack{m_1,m_2\\(m_1m_2,b)=1}}a_\xi(m_1)\overline{a_\xi(m_2)}\chi_{m_1}(kf^4)\overline{\chi_{m_2}(kf^4)}\\ \times W_0\Big(\frac{N(bfm_1)}{D}\Big)\overline{W_0\Big(\frac{N(bfm_2)}{D}\Big)}\widehat{\Phi}\Big(\frac{HN(k)}{N(f)^2N(m_1)N(m_2)}\Big).\end{aligned}\tag{4.16}$$

The characters enforce $(m_j,f)=1$, leaving only the displayed exclusion $(m_j,b)=1$. The supports of W_0 and $\widehat{\Phi}$ give

$$N(b)N(f)\le b_0D,\qquad 0<N(k)\le\frac{C_\Phi b_0^2D^2}{HN(b)^2}.$$

Partition $B\le N(b)<2B$ and $F\le N(f)<2F$ into dyadic ranges. Denote the corresponding contribution to (4.16) by $\mathcal{S}_{\xi;B,F}$, and put

$$X=\frac{D}{BF},\qquad \mathcal{H}=\frac{C_{I,\Phi}D^2}{HB^2},\qquad C_{I,\Phi}=\max(2,C_\Phi b_0^2).$$

These satisfy (4.11). In the variables $x_j=N(m_j)/X$, the coupled smooth weight is

$$\mathcal{K}_{b,f,k}(x_1,x_2)=W_0(rx_1)\overline{W_0(rx_2)}\widehat{\Phi}\Big(\frac{a}{x_1x_2}\Big),$$
$$r=\frac{N(b)N(f)}{BF}\in[1,4),\qquad a=\frac{HN(k)}{N(f)^2X^2}\le C_{I,\Phi}.$$

For every integer $q\ge0$, these kernels satisfy

$$\sup_{b,f,k}\|\mathcal{K}_{b,f,k}\|_{C^q([a_0/4,b_0]^2)}\ll_{I,\Phi,q}\|W\|^2_{C^q(I)}.$$

For $U\in C_c^\infty(I_*)$, where $I_*=[a_0/8,2b_0]$, put

$$S_U(b,f,k)=\sideset{}{^*}\sum_{\substack{(n,S)=1\\(n,b)=1}}a_\xi(n)\chi_n(k)\chi_n(f)^4U(N(n)/X).$$

The shifts $(X,F)\mapsto(X/N(d),FN(d))$ preserve (4.11). Thus Lemma 4.4 and (4.12), with $J=J(\varepsilon/4)$, give

$$\begin{aligned}\sum_{f,k}|S_U(b,f,k)|^2&=XF\,\mathcal{E}_{(b)}(\mathcal{H},X,F;\xi,U)\\&\le XF\,\tau_{\mathrm{div}}(b)\sum_{d|b}\mathcal{E}(\mathcal{H},X/N(d),FN(d);\xi,U)\\&\ll_{I,\varepsilon}D^{\varepsilon/2}(XF)^2\|U\|^2_{C^J(I_*)}.\end{aligned}$$

Here $N(b) \ll_I D$, so the divisor factors are absorbed in $D^{\varepsilon/4}$. We use the following smooth-weight principle, stated and proved in Lemma B.2: a mean-square bound valid for every common test function, with a C^J norm, also bounds the corresponding quadratic sum with a kernel depending on the row, at a cost given by the kernel's C^{2J+4} norm. For each fixed b, apply Lemma B.2 with row index (f,k), coefficient $\mu(f)\mathbf{1}_{(f,b)=1}$, and kernel $\mathcal{K}_{b,f,k}$. The larger row range $0 < N(k) \le \mathcal{H}$ adds only terms whose original kernel vanishes. With $q = 2J+4$, this gives

$$\begin{aligned} |\mathcal{S}_{\xi;B,F}| &\ll_{I,\varepsilon} D^{\varepsilon/2}\|W\|_{C^q(I)}^2 \sideset{}{^*}\sum_{B\le N(b)<2B} \frac{HN(b)}{D^2}(XF)^2 \\ &\ll D^{\varepsilon/2}\|W\|_{C^q(I)}^2 \frac{HB}{D^2}\, B\,(XF)^2 = D^{\varepsilon/2}\|W\|_{C^q(I)}^2 H, \end{aligned} \tag{4.17}$$

where $XF = D/B$ and ideal counting gives $O(B)$ choices of b. Summing the $O((\log D)^2)$ nonempty dyadic ranges bounds $\mathcal{S}_\xi$ as required. Sum over the fixed finite set of ξ, include Z, and restore the factor D to obtain (3.3). The required derivative order depends only on ε.

□

Remark 4.6. The formal proof organizes the dual mean squares slightly differently. The excluded primes are not removed by Lemma 4.4. Instead, an exclusion mask m is carried inside the dual mean square, through the twist $\chi_n(m^6) = \mathbf{1}_{(n,m)=1}$. Test functions are written as profiles $V(\log(\mathrm{N}_{K/\mathbb{Q}}(n)/X))$ with compact support and expanded once and for all in the height twists $\mathrm{N}_{K/\mathbb{Q}}(n)^{\mathrm{i}\theta}$, with bounds polynomial in θ. This is the expansion behind Lemma B.2, built into the definition. Finally, the rows are weighted by a compactly supported smooth majorant, whose Fourier transform is not compactly supported but decays rapidly, instead of a weight Φ with compactly supported $\widehat{\Phi}$; the resulting frequency tails are estimated separately. None of these changes affects the exponents.

5 Iteration of the dual mean-square estimate

Put $\Sigma = XF$. We prove the following estimate for the family (4.9), with exclusions removed by Lemma 4.4. The two inputs are proved in Sections 6 and 7.

Proposition 5.1. *Fix $\kappa > 0$ and $C_0 \ge 1$. Suppose that*

$$\mathcal{H}, X, F \ge 1, \qquad \Sigma = XF \le D^{C_0}, \qquad \mathcal{H} \le \Sigma D^{-\kappa}. \tag{5.1}$$

For every $\varepsilon > 0$ there is an integer $J = J(\kappa, C_0, \varepsilon) \ge 1$ such that

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) \ll_{I,\kappa,C_0,\varepsilon} \|W\|_{C^J(I)}^2 D^{\varepsilon}\Sigma \tag{5.2}$$

for every compact interval $I \subset (0,\infty)$ and smooth W supported in I.

The proof is given in Section 5.5, using Lemma 5.3 and Proposition 5.4.

5.1 The completed sums

Let Ψ be a completely multiplicative $\mathbb{C}$-valued function on the nonzero integral ideals of $\mathcal{O}_K = \mathbb{Z}[\omega]$ coprime to 3, vanishing on ideals divisible by a prime in S. For a primary element n, write $\Psi(n) = \Psi((n))$. Define

$$T(X;\Psi) = \sideset{}{^*}\sum_{(n,S)=1} \sum_{\substack{b\in\mathcal{O}\\ b\equiv 1 \ (3)\\ (b,S)=1}} \frac{\overline{\alpha(n)}\gamma_2(n)\Psi(n)\overline{\alpha(b)}^{\,3}\Psi(b)^3}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(n)}\,\mathrm{N}_{K/\mathbb{Q}}(b)}\, V_*(\mathrm{N}_{K/\mathbb{Q}}(n)\mathrm{N}_{K/\mathbb{Q}}(b)^3/X), \tag{5.3}$$

where $V_*(y) = \sqrt{y}\,W(y)$. The extra index b supplies the cubes in the Fourier expansion of the Kubota theta function. Here n, b are primary elements of $\mathcal{O}$; only n is required to be squarefree.

The $b = 1$ part is exactly $X^{-1/2}\sum^*_{(n,S)=1}\overline{\alpha(n)}\gamma_2(n)\Psi(n)W(\mathrm{N}_{K/\mathbb{Q}}(n)/X)$. To identify this with the normalized column sum in (4.9), fix a ray class character ξ. For a nonzero row $k \in \mathcal{O}$ and a squarefree primary f prime to S, set

$$\Psi_k(n) := \xi(n)\chi_n(k)\chi_n(f)^4, \qquad T(X;k,f) := T(X;\Psi_k). \tag{5.4}$$

The displayed product defines $\Psi_k(n)$ for $(n,S) = 1$; set $\Psi_k(n) = 0$ otherwise. Thus the zero extension required in (5.3) is part of this definition.

5.2 The completed mean-square estimate

For the twist (5.4), the theta transformation converts the relevant sextic twists into quadratic characters. Combining it with the quadratic large sieve gives the following estimate, proved in Section 6.

Proposition 5.2. *Fix $\varepsilon > 0$, $C_0 \geq 1$, and a character ξ of the fixed ray class group. There is $J = J(\varepsilon, C_0) \geq 1$ such that*

$$\sum_{0<\mathrm{N}_{K/\mathbb{Q}}(k)\ll\mathcal{H}} |T(X;k,f)|^2 \ll_{I,\xi,\varepsilon,C_0} D^{\varepsilon}\|W\|^2_{C^J(I)}\Big(\mathcal{H} + \frac{\mathcal{H}^2\mathrm{N}_{K/\mathbb{Q}}(f)}{X}\Big) \tag{5.5}$$

whenever

$$1 \leq \mathcal{H}, X, \mathrm{N}_{K/\mathbb{Q}}(f) \leq D^{C_0}.$$

Here f is squarefree and primary with $(f,S) = 1$, W is smooth and supported in a compact interval $I \subset (0,\infty)$, and T is defined by (5.4).

5.3 Cube inversion and the remaining sums

Set

$$H_c^3 = \min\Big(X, \frac{X^2}{\mathcal{H}^2}\Big), \qquad L_b = \frac{X}{\mathrm{N}_{K/\mathbb{Q}}(b)^3}. \tag{5.6}$$

Lemma 5.3. *Fix $C_0 \geq 1$ and $\varepsilon > 0$. Suppose $1 \leq \mathcal{H}, X, F \leq D^{C_0}$ and $\mathcal{H} \leq \Sigma = XF$. There is an integer $J = J(\varepsilon, C_0) \geq 1$ such that*

$$\mathcal{E}(\mathcal{H},X,F;\xi,W) \ll_{I,C_0,\varepsilon} D^{\varepsilon}\Bigg(\Sigma\|W\|^2_{C^J(I)} + \sup_{\substack{b\equiv 1\ (3),\ (b,S)=1\\ \mathrm{N}_{K/\mathbb{Q}}(b)>H_c,\ L_b>1}} \mathcal{E}(\mathcal{H},L_b,F;\xi,W)\Bigg) \tag{5.7}$$

for every compact interval $I \subset (0,\infty)$ and $W \in C_c^\infty(I)$. An empty supremum is zero; in particular it is empty when $\mathcal{H}^2 \leq X$.

Proof. Write $N = \mathrm{N}_{K/\mathbb{Q}}$ and $I = [u,v]$. Complete multiplicativity in (5.3), including at the zeros of Ψ_k, gives

$$\begin{aligned} X^{-1/2}\sum^*_{(n,S)=1} a_\xi(n)\chi_n(k)\chi_n(f)^4 W(N(n)/X) \\ = \sum_{(h,S)=1} \frac{\mu(h)\overline{\alpha(h)}^3\Psi_k(h)^3}{N(h)} T(X/N(h)^3;k,f). \end{aligned} \tag{5.8}$$

Indeed, after substitution of (5.3), the coefficient of the total cube index b contains $\sum_{h|b}\mu(h) = \mathbf{1}_{b=1}$. This is Möbius inversion; compare [14, (1.18)] and [5, (8.2)].

Split the right-hand side of (5.8) into $P_{\mathrm{short}}(k,f)+P_{\mathrm{long}}(k,f)$, where

$$P_{\mathrm{short}}(k,f)=\sum_{\substack{(h,S)=1\\ N(h)\le H_c}}\frac{\mu(h)\overline{\alpha(h)}^3\Psi_k(h)^3}{N(h)}T(X/N(h)^3;k,f),$$

$$\mathcal{E}_{\mathrm{short}}=\frac{1}{F}\sideset{}{^*}\sum_{\substack{F\le N(f)<2F\\ (f,S)=1}}\sum_{0<N(k)\le\mathcal{H}}|P_{\mathrm{short}}(k,f)|^2.$$

Define $\mathcal{E}_{\mathrm{long}}$ in the same way using P_{long}. Weighted Cauchy–Schwarz for each k,f, followed by Proposition 5.2, gives

$$\begin{aligned}\mathcal{E}_{\mathrm{short}}&\le\Big(\sum_{N(h)\le H_c}\frac{1}{N(h)}\Big)\sum_{N(h)\le H_c}\frac{1}{N(h)}\frac{1}{F}\sideset{}{^*}\sum_{\substack{F\le N(f)<2F\\ (f,S)=1}}\sum_{0<N(k)\le\mathcal{H}}|T(X/N(h)^3;k,f)|^2\\ &\ll_{I,C_0,\varepsilon}D^{\varepsilon/2}\|W\|^2_{C^J(I)}\Big(\mathcal{H}+\frac{\mathcal{H}^2FH_c^3}{X}\Big)\ll D^{\varepsilon/2}\Sigma\|W\|^2_{C^J(I)}.\end{aligned}\tag{5.9}$$

Here $X/N(h)^3\ge 1$, and the reciprocal-norm sum contributes only a logarithm. We use Proposition 5.2 with exponent $\varepsilon/4$ and range C_0+1, since $N(f)<2D^{C_0}$. If $H_c<1$, the sum is empty.

Expand T using (5.3) in the remaining terms, with cube index c:

$$\begin{aligned}P_{\mathrm{long}}(k,f)=\frac{1}{\sqrt{X}}\sum_{\substack{(h,S)=1\\ N(h)>H_c}}\sum_{(c,S)=1}&\mu(h)\sqrt{N(hc)}\,\overline{\alpha(hc)}^3\Psi_k(hc)^3\\ &\times\sideset{}{^*}\sum_{(n,S)=1}a_\xi(n)\chi_n(k)\chi_n(f)^4W\Big(\frac{N(n)N(hc)^3}{X}\Big).\end{aligned}$$

Put $b=hc$; the pairs giving b are exactly $(h,b/h)$ with $h\mid b$ and $N(h)>H_c$. Since $\Psi_k(b)^3=\xi(b)^3\chi_b(k)^3\mathbf{1}_{(b,f)=1}$, this gives

$$P_{\mathrm{long}}(k,f)=\sum_{\substack{(b,S)=1\\ N(b)>H_c}}\frac{\beta_0(b,f)\chi_b(k)^3}{N(b)}\frac{1}{\sqrt{L_b}}\sideset{}{^*}\sum_{(n,S)=1}a_\xi(n)\chi_n(k)\chi_n(f)^4W(N(n)/L_b),$$

$$\beta_0(b,f):=\overline{\alpha(b)}^3\xi(b)^3\mathbf{1}_{(b,f)=1}\sum_{\substack{h\mid b\\ N(h)>H_c}}\mu(h),\qquad|\beta_0(b,f)|\le\tau_{\mathrm{div}}(b).$$

The divisor function $\tau_{\mathrm{div}}(b)$ counts ideal divisors. All these indices are primary and prime to S; b need not be squarefree. The support of W restricts the sum to $N(b)^3\le vX$. Weighted Cauchy–Schwarz now gives

$$\mathcal{E}_{\mathrm{long}}\le\Big(\sum_{\substack{H_c<N(b)\le(vX)^{1/3}\\ (b,S)=1}}\frac{\tau_{\mathrm{div}}(b)}{N(b)}\Big)\sum_{\substack{H_c<N(b)\le(vX)^{1/3}\\ (b,S)=1}}\frac{\tau_{\mathrm{div}}(b)}{N(b)}\mathcal{E}(\mathcal{H},L_b,F;\xi,W).$$

The reciprocal divisor sum is $\ll_I\log^2(2+X)$, since

$$\sum_{N(b)\le R}\frac{\tau_{\mathrm{div}}(b)}{N(b)}=\sum_{N(cd)\le R}\frac{1}{N(c)N(d)}\le\Big(\sum_{N(c)\le R}\frac{1}{N(c)}\Big)^2\ll\log^2(2R)\qquad(R\ge1).$$

If $L_b \le 1$, the column sum is empty unless $L_b \ge 1/v$; otherwise it has $O_I(1)$ terms, so

$$\mathcal{E}(\mathcal{H}, L_b, F; \xi, W) \ll_I \frac{\mathcal{H}}{L_b}\|W\|_\infty^2 \ll_I \Sigma\|W\|_\infty^2.$$

Absorb the logarithms in D^ε and combine the two parts using $\mathcal{E} \le 2\mathcal{E}_{\mathrm{short}} + 2\mathcal{E}_{\mathrm{long}}$. This proves (5.7). If $\mathcal{H}^2 \le X$, then $H_c = X^{1/3}$, so $N(b) > H_c$ implies $L_b < 1$. □

5.4 The transfer estimate

Fix a nonnegative radial Schwartz weight Φ with $\Phi(t) \ge 1$ for $0 \le t \le 1$ and $\operatorname{supp}\widehat{\Phi} \subset [0, C_\Phi]$. Such a weight is obtained by squaring and rescaling a real radial Schwartz function with compactly supported Fourier transform. Write $N = \mathrm{N}_{K/\mathbb{Q}}$. For fixed $\mathcal{H}, L, F, \xi$, put

$$\mathcal{A}(W) := \frac{1}{LF} \sideset{}{^*}\sum_{\substack{F \le N(f) < 2F \\ (f,S)=1}} \sum_{k \in \mathcal{O}} \Phi(N(k)/\mathcal{H}) \Big| \sideset{}{^*}\sum_{(n,S)=1} a_\xi(n)\chi_n(k)\chi_n(f)^4 W(N(n)/L)\Big|^2. \tag{5.10}$$

Thus $\mathcal{E}(\mathcal{H}, L, F; \xi, W) \le \mathcal{A}(W)$. In the following proposition Σ is an independent target scale; we will apply it at column scale L_b with $\Sigma = XF$.

Proposition 5.4. *Assume $1 \le \mathcal{H}, L, F, \Sigma \le D^{C_0}$ and $\max\{\mathcal{H}, LF\} \le \Sigma$, for fixed $C_0 \ge 1$. For every integer $m \ge 0$ and $\varepsilon > 0$,*

$$\mathcal{A}(W) \ll D^\varepsilon \Sigma \|W\|_{C^{4m+12}(I)}^2 \Big(1 + \sup \frac{\mathcal{E}(\mathcal{H}', X', F'; \xi', U)}{\Sigma'}\Big) \tag{5.11}$$

where the supremum is over the family (4.9), with $\Sigma' = X'F'$, $\mathcal{H}', X', F' \ge 1$, and

$$\mathcal{H}' \le \frac{\mathcal{H}L}{\Sigma F}, \qquad \frac{\mathcal{H}'}{\Sigma'} \le \frac{\mathcal{H}}{\Sigma}, \qquad \Sigma' \le L. \tag{5.12}$$

For $I = [u, v]$, the tests satisfy $U \in C_c^\infty(I')$, $I' = [u/16, 4v]$, and $\|U\|_{C^m(I')} \le 1$; an empty supremum is zero. The estimate holds for every compact $I \subset (0, \infty)$ and $W \in C_c^\infty(I)$, with implied constant depending on m, C_0, ε, I and the fixed ray class group.

The proof is given in Section 7, by combining Lemmas 7.1 and 7.3.

5.5 Proof of Proposition 5.1

Proof of Proposition 5.1. Fix $\kappa > 0$ and $C_0 \ge 1$. We prove by induction on $j \ge 0$ that the proposition holds under the additional restriction $\mathcal{H} \le D^{j\kappa}$, with the derivative order and implied constant allowed to depend on j. More precisely, for every $\varepsilon > 0$ there is an integer $J = J(j, \kappa, C_0, \varepsilon) \ge 1$ such that

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) \ll_{I,j,\kappa,C_0,\varepsilon} D^\varepsilon \Sigma \|W\|_{C^J(I)}^2$$

for every compact interval $I = [u, v] \subset (0, \infty)$ and $W \in C_c^\infty(I)$, under the hypotheses of the proposition. The derivative order is independent of I; this allows us to apply the induction hypothesis on the enlarged interval in the transfer estimate. For the base case $j = 0$, we have $\mathcal{H} \le 1$. In (4.9), the support of W restricts n to the $O_I(X)$ ideals with $\mathrm{N}_{K/\mathbb{Q}}(n) \in [uX, vX]$. The factors $a_\xi(n)$, $\chi_n(k)$, and $\chi_n(f)^4$ have absolute value at most one, so the triangle inequality bounds each inner sum by $O_I(X\|W\|_\infty)$. There are $O(F)$ choices of f and $O(\mathcal{H})$ choices of k. Consequently,

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) \ll_I \frac{F\mathcal{H}}{XF}\big(X\|W\|_\infty\big)^2 = \mathcal{H}X\|W\|_\infty^2.$$

Dividing by $\Sigma = XF$ and using $\mathcal{H} \le 1$ and $F \ge 1$ proves the assertion for $j = 0$, with $J = 1$.

Suppose the assertion holds for j, and fix $\varepsilon > 0$. Let m be the derivative order supplied by the induction hypothesis with exponent $\varepsilon/3$. For $\mathcal{H} \le D^{(j+1)\kappa}$, apply Lemma 5.3 with exponent $\varepsilon/3$. We must bound the mean squares in its supremum uniformly in b. If this supremum is nonempty, then $\mathcal{H}^2 > X$ and $H_c^3 = X^2/\mathcal{H}^2$. For each such b, recall that $L_b = X/\mathrm{N}_{K/\mathbb{Q}}(b)^3 > 1$. Since $L_b \le X$ and $\Sigma \ge \max\{\mathcal{H}, L_b F\}$, Proposition 5.4 applies at column scale L_b, with derivative order m, and exponent $\varepsilon/3$. Its estimate also bounds $\mathcal{E}(\mathcal{H}, L_b, F; \xi, W)$ because $\mathcal{E}(\mathcal{H}, L_b, F; \xi, W) \le \mathcal{A}(W)$.

Let $\mathcal{H}', X', F'$ be any parameters in the supremum in (5.11), with $\Sigma' = X'F'$ as in that proposition. By (5.12) and $\mathrm{N}_{K/\mathbb{Q}}(b) > H_c$,

$$\mathcal{H}' \le \frac{\mathcal{H} L_b}{\Sigma F} = \frac{\mathcal{H}}{\mathrm{N}_{K/\mathbb{Q}}(b)^3 F^2} < \mathcal{H}\Big(\frac{\mathcal{H}}{\Sigma}\Big)^2 \le D^{-2\kappa}\mathcal{H} \le D^{j\kappa},$$
$$\frac{\mathcal{H}'}{\Sigma'} \le \frac{\mathcal{H}}{\Sigma} \le D^{-\kappa}, \qquad \Sigma' \le L_b \le \Sigma \le D^{C_0}.$$

Thus each of these mean squares is covered by the induction hypothesis. Its weight U is supported in $[u/16, 4v]$ and has C^m norm at most one. Applying the induction hypothesis on this interval gives

$$\frac{\mathcal{E}(\mathcal{H}', X', F'; \xi', U)}{\Sigma'} \ll_{I,j,\kappa,C_0,\varepsilon} D^{\varepsilon/3}.$$

The enlarged interval is determined by I, so the implied constant has only the permitted dependence on the support. Substituting into (5.11) yields, uniformly in b,

$$\mathcal{E}(\mathcal{H}, L_b, F; \xi, W) \ll_{I,j,\kappa,C_0,\varepsilon} D^{2\varepsilon/3}\Sigma \|W\|^2_{C^{4m+12}(I)}.$$

Substitution into (5.7) contributes the remaining factor $D^{\varepsilon/3}$. Choose J at least $4m + 12$ and at least the derivative order required by Lemma 5.3. We obtain

$$\mathcal{E}(\mathcal{H}, X, F; \xi, W) \ll_{I,j,\kappa,C_0,\varepsilon} D^{\varepsilon}\Sigma \|W\|^2_{C^J(I)}.$$

The choice of J depends only on $j, \kappa, C_0, \varepsilon$. If the supremum in (5.7) is empty, the same bound follows directly from that lemma. This completes the induction.

Finally, take $j = \lceil C_0/\kappa \rceil$. The hypothesis $\mathcal{H} \le \Sigma \le D^{C_0}$ ensures $\mathcal{H} \le D^{j\kappa}$, so the induction gives the proposition. □

Remark 5.5. The formal proof runs the same descent with fixed parameters. At $\kappa = 1/20$ (the value used in Section 5.6) it removes, at each level, the cubes of norm greater than a fixed cutoff $D^{1/1000}$, instead of the adaptive cutoff H_c of (5.6). Each level therefore lowers the row norm by a factor of at least $D^{1/1000}$ (in fact $D^{3/1000}$), so the rank $\lceil 1000 \log \mathcal{H}/\log D \rceil$ drops at each level. Since $\mathcal{H} \le D^3$ in all states, at most 3000 levels occur, and the induction is on this rank rather than on j. The losses are tracked additively: each level costs a factor $D^{57\eta}$, and the gap $1/20$ is reduced by η per level, with $3000\eta \le 1/80$, instead of re-quantifying ε at each level. The short cubes then contribute $D^{7\varepsilon - 1/40}\Sigma$: the saving $D^{-1/40}$, half the gap, absorbs the loss $D^{7\varepsilon}$. Because the margin starts at $1/20$, the formal mean square requires $\vartheta > 1/20$, which $\vartheta = 1/10$ satisfies.

5.6 The original mean square

Proof of Proposition 3.1. Let $\vartheta = 1/10$ and put $H = D^{1+\vartheta} = D^{11/10}$. The ranges in (4.11) give

$$\mathcal{H} \le \frac{CD^{1-\vartheta}}{B^2}, \qquad \Sigma = XF = \frac{D}{B}, \qquad \frac{\mathcal{H}}{\Sigma} \ll D^{-\vartheta}. \tag{5.13}$$

For large D, Proposition 5.1 applies with $\kappa = \vartheta/2 = 1/20$ and $C_0 = 2$. If $X < 1$, nonempty support forces $X \gg_I 1$, and counting gives $\mathcal{E} \ll_I \mathcal{H}\|W\|_\infty^2 \ll \Sigma\|W\|_\infty^2$; if $\mathcal{H} < 1$, the sum is empty. Thus (4.12) holds, with derivative order depending only on ε, and Proposition 4.5 proves (3.3). Bounded D is again covered by counting. □

6 Proof of the completed mean-square estimate

We prove Proposition 5.2. We first express the completed sum (5.3) using cubic theta coefficients and state the transformation formula. We then apply the quadratic large sieve and account for repeated prime factors in k.

6.1 Realization by the cubic theta function

With the row k and auxiliary index f fixed, we express $T(X;k,f)$ as a weighted sum of Fourier coefficients of the cubic theta function. Its automorphy then expresses this sum in terms of coefficients at other cusps, as in [5, §5 and Appendix A]. We use Ψ_k from (5.4), suppressing its dependence on the fixed f and ξ.

Write $(z,v) \in \mathbb{C} \times \mathbb{R}_{>0}$ for upper half-space coordinates, with horizontal coordinate z and height v. We use Kubota's cubic theta function in the normalization of [5, §5.1, (5.6)]:

$$\theta(z,v) = \frac{3^{5/2}}{2} v^{2/3} + \sum_{\substack{\ell \in \lambda^{-3}\mathcal{O} \\ \ell \neq 0}} \tau(\ell) v K_{1/3}(4\pi|\ell|v) \exp\bigl(2\pi\mathrm{i}(\ell z + \overline{\ell z})\bigr). \tag{6.1}$$

Here $K_{1/3}$ is the modified Bessel function of the second kind, and $\tau(\ell)$ is the coefficient sequence given explicitly in [5, (5.7)–(5.8)], following Patterson's calculation [23, Theorem 8.1].

We define $\Theta_k(z,v)$ by twisting the Fourier coefficients of $\overline{\theta}$. First define $\phi_k : \mathcal{O} \to \mathbb{C}$ by

$$\phi_k(n) = \begin{cases} \chi_n(\lambda)^2 \Psi_k(n), & n \equiv 1 \pmod 3, \quad (n,S) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then we define $\Theta_k(z,v)$ by multiplying the Fourier coefficient of $\overline{\theta}$ at ℓ, which is $\overline{\tau(-\ell)}$, by $\phi_k(\lambda^3\ell)$:

$$\Theta_k(z,v) = \sum_{\substack{\ell \in \lambda^{-3}\mathcal{O} \\ \ell \neq 0}} \overline{\tau(-\ell)} \phi_k(\lambda^3\ell) v K_{1/3}(4\pi|\ell|v) \exp\bigl(2\pi\mathrm{i}(\ell z + \overline{\ell z})\bigr). \tag{6.2}$$

Recall that $c_\theta(nb^3)$ is defined in (2.9). Choose a nonzero $q \in \mathcal{O}$ such that ϕ_k is periodic modulo (q), and put

$$\widehat{\phi}_k(h) = \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(q)} \sum_{n \bmod q} \phi_k(n) e(-hn/q).$$

Lemma 6.1. *With these definitions,*

$$T(X;k,f) = \frac{1}{3^{5/2}\sqrt{X}} \sum_{\substack{n,b \in \mathcal{O} \\ n,b \equiv 1 \ (3) \\ n \text{ squarefree} \\ (nb,S)=1}} c_\theta(nb^3) \phi_k(nb^3) \overline{\alpha(nb^3)} W\Bigl(\frac{\mathrm{N}_{K/\mathbb{Q}}(nb^3)}{X}\Bigr). \tag{6.3}$$

Moreover,

$$\Theta_k(z,v) = \sum_{h \in \mathcal{O}/(q)} \widehat{\phi}_k(h) \overline{\theta(z + \lambda^2 h/q, v)}. \tag{6.4}$$

Proof. The coefficient formula (2.9) gives

$$c_\theta(nb^3)\phi_k(nb^3) = 3^{5/2}|b|\gamma_2(n)\Psi_k(n)\Psi_k(b)^3.$$

Substitution into (5.3), using $V_*(y) = \sqrt{y}\,W(y)$, proves (6.3). Finite Fourier inversion gives

$$\phi_k(n) = \sum_{h\in\mathcal{O}/(q)} \widehat{\phi}_k(h)e(nh/q), \qquad \sum_{h\in\mathcal{O}/(q)} \widehat{\phi}_k(h) = \phi_k(0) = 0.$$

Multiplication of the ℓth Fourier mode by $e(\lambda^3\ell h/q)$ translates z to $z + \lambda^2 h/q$. The second identity cancels the constant term, proving (6.4). □

The resulting transformation formula is stated in the next subsection; its automorphy calculation is given in Appendix A.2.

6.2 The theta transformation

Recall that, for a primary prime $p \notin S$, $\chi_p(x) = (x/p)_6$ is the sextic residue character on $(\mathcal{O}/(p))^\times$. For every integer j, write χ_p^j for its jth power on this group, extended by zero on multiples of p; in particular, $\chi_p^0(x) = \mathbf{1}_{p\nmid x}$. Sextic reciprocity (4.1) expresses the factors of Ψ_k at primes outside S as such powers, with $0 \le j \le 5$.

For a primary prime $p \notin S$, $j \in \{0,\dots,5\}$, and $x \in \mathcal{O}$, define the local factor

$$B_{p,j}(x) = \begin{cases} \chi_p(x)^{-j-2}, & j \neq 0, 4, \\ \mathrm{N}_{K/\mathbb{Q}}(p)^{-1/2}(-1 + \mathrm{N}_{K/\mathbb{Q}}(p)\mathbf{1}_{p|x}), & j = 4, \\ \mathrm{N}_{K/\mathbb{Q}}(p)^{-1/2}\chi_p(x)^{-2}, & j = 0. \end{cases} \tag{6.5}$$

The key case is the quadratic character

$$B_{p,1}(x) = \chi_p(x)^3.$$

For the smooth compactly supported weight V_* in (5.3), put $\widehat{V}_*(s) = \int_0^\infty V_*(x)x^s\,dx/x$. We write $\int_{(\sigma)}$ for integration upwards along the vertical line with real part σ. With Γ denoting Euler's gamma function, the accompanying transform of the weight is

$$V_*^\sharp(x) = \frac{1}{2\pi\mathrm{i}}\int_{(0)} \widehat{V}_*(-t)\frac{\Gamma(7/6+t)\Gamma(5/6+t)}{\Gamma(7/6-t)\Gamma(5/6-t)}\Big(\frac{(2\pi)^4 x}{27}\Big)^{-t}\,dt, \qquad x > 0. \tag{6.6}$$

Here θ is Kubota's cubic theta function, normalized in (6.1). The dual sums use the Fourier coefficients of $\overline{\theta}$ at three cusps. Take the representatives

$$\gamma_0 = I, \qquad \gamma_+ = \begin{pmatrix} 1 & 0 \\ \omega & 1 \end{pmatrix}, \qquad \gamma_- = \begin{pmatrix} 1 & 0 \\ \omega^2 & 1 \end{pmatrix}, \tag{6.7}$$

acting on upper half-space. For $\sigma \in \{0, +, -\}$, define $d_\sigma(\ell)$ by the Fourier expansion [5, (5.9), (5.15)]

$$\overline{\theta(\gamma_\sigma(z,v))} = \frac{3^{5/2}}{2}\mathbf{1}_{\sigma=0}v^{2/3} + \sum_{0\neq\ell\in\lambda^{-4}\mathcal{O}} d_\sigma(\ell)vK_{1/3}(4\pi|\ell|v)\exp\big(2\pi\mathrm{i}(\ell z + \overline{\ell z})\big). \tag{6.8}$$

Thus $d_0(\ell) = \overline{\tau(-\ell)}$, with τ extended by zero outside $\lambda^{-3}\mathcal{O}$. In particular, $c_\theta(nb^3) = d_0(\lambda^{-3}nb^3)$ for the indices in (2.9); the outline uses $d_\theta(m)$ for $d_\sigma(\lambda^{-4}m)$ with one of these three choices of σ. The arithmetic formulas for all three sequences are given by (A.6) and (A.7) in Appendix A.2.

We state the formula for a general product of local twists. Fix a ray class character whose conductor is supported on S, and let Ψ_0 be its extension by zero at every prime in S. Let $\mathcal{P}$

be a finite set of primes outside S, each represented by its primary generator in $\mathcal{O}$, and choose integers $0 \le j_p \le 5$ for each $p \in \mathcal{P}$. For primary $n \in \mathcal{O}$, set

$$\Psi(n) = \Psi_0(n) \prod_{p \in \mathcal{P}} \chi_p^{j_p}(n).$$

In the transformed sums, $\mathcal{A}$ will range over subsets satisfying

$$\{p \in \mathcal{P} : j_p \neq 0\} \subseteq \mathcal{A} \subseteq \mathcal{P}.$$

Thus only primes with $j_p = 0$ may be omitted from $\mathcal{A}$. We call the primes in $\mathcal{A}$ *active* and those in $\mathcal{P} \setminus \mathcal{A}$ *inactive*. For such a subset and $c_0 \in \mathcal{O} \setminus \{0\}$, write

$$c = c_0 \prod_{p \in \mathcal{A}} p.$$

Proposition 6.2. *The quantity $T(X;\Psi)$ defined in (5.3) is a sum of $O_{\Psi_0}(2^{|\mathcal{P}|})$ terms of the form*

$$C \sum_{0 \neq \ell \in \lambda^{-4}\mathcal{O}} \frac{d(\ell)\alpha(\ell)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(\ell)}} \psi(\lambda^4 \ell) \prod_{p \in \mathcal{A}} B_{p,j_p}(\lambda^4 \ell) V_*^\sharp\Big(\frac{\mathrm{N}_{K/\mathbb{Q}}(\ell) X}{\mathrm{N}_{K/\mathbb{Q}}(c)^2}\Big). \tag{6.9}$$

Here $\mathcal{A}$ and c are as above, and $|C| \ll_{\Psi_0} 1$. The triples (d, ψ, c_0) belong to a fixed finite family depending only on Ψ_0, where $d \in \{d_0, d_+, d_-\}$, $c_0 \in \mathcal{O} \setminus \{0\}$, and ψ is a unit-modulus additive character on $\mathcal{O}$.

To average the transformed sums over k_0, we need to choose their cusp coefficients and additive characters consistently as k_0 varies. Fix a finite set $\mathcal{P}_{\mathrm{fix}}$ of primary primes outside S and exponents $j_p \in \{0, \dots, 5\}$, and put

$$\Psi_{k_0}(n) = \Psi_0(n) \prod_{p \in \mathcal{P}_{\mathrm{fix}}} \chi_p^{j_p}(n) \prod_{p \mid k_0} \chi_p(n),$$

where k_0 is primary and squarefree, with $(k_0, S\prod_{p \in \mathcal{P}_{\mathrm{fix}}} p) = 1$. Let $\mathcal{A}_{\mathrm{fix}}$ range over the subsets satisfying

$$\{p \in \mathcal{P}_{\mathrm{fix}} : j_p \neq 0\} \subseteq \mathcal{A}_{\mathrm{fix}} \subseteq \mathcal{P}_{\mathrm{fix}}, \qquad \mathcal{A} = \mathcal{A}_{\mathrm{fix}} \cup \{p : p \mid k_0\}.$$

Every prime dividing k_0 belongs to $\mathcal{A}$, since its exponent is 1.

Lemma 6.3 (Uniformity in the twist)**.** *For the family Ψ_{k_0} above, the summands in Proposition 6.2 may be indexed by $(h, \mathcal{A}_{\mathrm{fix}})$, with h in a fixed finite set depending only on Ψ_0. Zero scalar coefficients are permitted. For each fixed index, the triple (d, ψ, c_0) depends on k_0 only through its ray class modulo a fixed ideal supported on S and depending only on Ψ_0.*

To estimate the dual sums, we need bounds for the cusp coefficients and the transformed weight.

Lemma 6.4. *For each $d \in \{d_0, d_+, d_-\}$, the coefficient $d(\ell)$ vanishes unless ℓ can be written as*

$$\ell = u\lambda^m n b^3, \tag{6.10}$$

where $u \in \mathcal{O}^\times$, $m \in \mathbb{Z}$ with $m \ge -4$, and $n, b \in \mathcal{O}$ are primary with n squarefree.

$$|d(\ell)| = |d(u\lambda^m n b^3)| \le 27 \cdot 3^{m/6} |b|. \tag{6.11}$$

For $A > 0$, integers $j \ge 0$, and V_* supported in a fixed compact interval $I \subset (0, \infty)$, there is $J = J(A, j)$ such that

$$|(x\partial_x)^j V_*^\sharp(x)| \ll_{A,j,I} \|V_*\|_{C^J(I)} \min(x^{1/4}, x^{-A}), \qquad x > 0. \tag{6.12}$$

Proposition 6.2 and Lemmas 6.3 and 6.4 are proved in Appendix A.2.

6.3 The quadratic mean square on the dual side

We record the quadratic large-sieve estimate needed below.

Lemma 6.5. *Let $\mathcal{H}, U \geq 1$, and let $\beta(n)$ be complex coefficients on squarefree primary n with $\mathrm{N}_{K/\mathbb{Q}}(n) \asymp U$. For every $\varepsilon > 0$,*

$$\sum_{\substack{k \equiv 1 \ (3),\ (k,S)=1 \\ \mathrm{N}_{K/\mathbb{Q}}(k) \leq \mathcal{H}}}^{*} \Big| \sum_{\substack{n \equiv 1 \ (3),\ n \text{ squarefree} \\ \mathrm{N}_{K/\mathbb{Q}}(n) \asymp U}} \beta(n) \chi_k(n)^3 \Big|^2 \ll_\varepsilon (\mathcal{H}U)^\varepsilon (\mathcal{H} + U) \sum_n |\beta(n)|^2. \tag{6.13}$$

The star restricts k to squarefree primary elements.

Proof. This is Goldmakher and Louvel's quadratic large sieve [7, Theorem 1.1], after fixing the product of the prime factors of n lying in S, and finitely many ray classes. For completeness, let $e_k \in \{0, 1\}$ according as $\mathrm{N}_{K/\mathbb{Q}}(k) \equiv 1, 3 \pmod 4$, and let κ_λ be the nontrivial character modulo λ. The character $x \mapsto (x/k)_2 \kappa_\lambda(x)^{e_k}$ is trivial on units and has primitive conductor $k\lambda^{e_k}$. When $e_k = 0$, the factor $\kappa_\lambda(x)^{e_k}$ is omitted. Classes modulo $24\mathcal{O}$ fix the supplementary characters and reciprocity factors; for coprime k_1, k_2 in the same class, the product character has conductor $k_1 k_2$. These are the hypotheses in [7, Definition 1 and §2]. □

6.4 Squarefree rows and the completed bound

Write $N(a) = \mathrm{N}_{K/\mathbb{Q}}(a)$. In (5.4), allow any $g \in \mathcal{O} \setminus \{0\}$ in place of f, so $\Psi_k(n) = \xi(n) \chi_n(k) \chi_n(g)^4$. The zero extension at S is retained. Choose prime-ideal generators, primary away from 3, and extend multiplicatively to all ideals. We first bound squarefree rows.

Lemma 6.6. *For every $\varepsilon > 0$ and $C_0 \geq 1$ there is $J = J(\varepsilon, C_0) \geq 1$ such that*

$$\sum_{\substack{0 < N(s) \leq \mathcal{H} \\ s \text{ squarefree}}} |T(X; u_0 s, g)|^2 \ll_{I, \xi, \varepsilon, C_0} D^\varepsilon \|W\|_{C^J(I)}^2 \Big(\mathcal{H} + \frac{\mathcal{H}^2 N(g)}{X} \Big) \tag{6.14}$$

for $1 \leq \mathcal{H}, X, N(g) \leq D^{C_0}$, $u_0 \in \mathcal{O}^\times$, every compact interval $I \subset (0, \infty)$, and $W \in C_c^\infty(I)$. The sum uses the chosen generators of squarefree ideals, including those meeting S.

Proof. By homogeneity assume $\|W\|_{C^J(I)} \leq 1$, with J chosen below. Write

$$s = tk_0, \qquad t = \prod_{\substack{p \mid s \\ p \mid g \text{ or } p \in S}} p, \qquad N(k_0) \leq \mathcal{H}_0 := \frac{\mathcal{H}}{N(t)}.$$

Fix t; then k_0 is squarefree and primary, with $(k_0, g) = 1$ and $(k_0, S) = 1$. Discard empty ranges, so $\mathcal{H}_0 \geq 1$, and retain these restrictions below. For a coefficient function $A(n, b)$, a positive scale Y, and an integer $m \geq -4$, put

$$\mathscr{S}_m[A, Y](k_0) := \sum_{\substack{n, b \equiv 1 \ (3) \\ n \text{ squarefree}}} \frac{A(n, b) \chi_{k_0}(nb)^3}{3^{m/3} \sqrt{N(n)}\, N(b)} V_*^\sharp\Big(\frac{3^m N(n) N(b)^3 \mathcal{H}_0^2}{Y N(k_0)^2} \Big). \tag{6.15}$$

We shall obtain, after partitioning k_0 into fixed ray classes,

$$T(X; u_0 t k_0, g) = \sum_{\iota \in \mathcal{I}} \sum_{m \geq -4} c_{\iota, m}(k_0) \mathscr{S}_m[A_{\iota, m}, Y_\iota](k_0),$$

where $\mathcal{I}$ is independent of k_0, $|\mathcal{I}| \preccurlyeq 1$, $|c_{\iota,m}(k_0)| \ll 1$, and $A_{\iota,m}$ is independent of k_0. For some $a_\iota, Y_\iota > 0$, the coefficients and lengths satisfy

$$|A_{\iota,m}(n,b)| \ll a_\iota, \qquad a_\iota^2 \le 1, \qquad a_\iota^2 Y_\iota \ll \frac{\mathcal{H}_0^2}{X} N(t)^2 N(g). \tag{6.16}$$

All column restrictions are included by extending $A_{\iota,m}$ by zero. Put $k = u_0 t k_0$. For $\mathcal{P} = \{p \notin S : p \mid kg\}$, sextic reciprocity (4.1) gives

$$\Psi_k(n) = \Psi_0(n) \prod_{p \in \mathcal{P}} \chi_p^{j_p}(n), \qquad j_p \equiv v_p(k) + 4v_p(g) \pmod 6, \qquad 0 \le j_p \le 5. \tag{6.17}$$

Even when $j_p = 0$, the factor $\chi_p^0(n) = \mathbf{1}_{p \nmid n}$ retains the zero extension at $p \mid kg$. The fixed factor Ψ_0 contains ξ, the factors at S, the unit factors, and $n \mapsto \mathcal{R}(n, \prod_{p \notin S} p^{v_p(k)})$. The fourth power at g contributes no reciprocity sign, and the factors at S depend only on exponents modulo six. Thus Ψ_0 ranges over a fixed finite family. In the notation of Lemma 6.3, take $\mathcal{P}_{\mathrm{fix}} = \{p \notin S : p \mid tg\}$, so that $\mathcal{P} = \mathcal{P}_{\mathrm{fix}} \sqcup \{p : p \mid k_0\}$; the exponents on $\mathcal{P}_{\mathrm{fix}}$ are fixed with t, g. Partitioning k_0 into fixed ray classes fixes Ψ_0 and, by Lemma 6.3, the data d, ψ, c_0 in each transformed term.

Fix one transformed term, with active primes $\mathcal{A}_{\mathrm{fix}}$ away from k_0, and put $c_* = c_0 \prod_{p \in \mathcal{A}_{\mathrm{fix}}} p$. Thus $c = c_* k_0$ in (6.9). By (6.10), write $\ell = u\lambda^m n b^3$, with u, m, n, b as there. Every prime dividing k_0 has $j_p = 1$, and hence

$$B_{p,1}(u\lambda^{m+4} n b^3) = \chi_p(u\lambda^{m+4})^3 \chi_p(nb)^3. \tag{6.18}$$

Indeed, the transformed exponent is $-1-2 \equiv 3 \pmod 6$ and $\chi_p(b)^9 = \chi_p(b)^3$, also when $p \mid b$ by zero extension. The coefficient bound and rapid decay in Lemma 6.4 give absolute convergence of the transformed series, so we may regroup its terms below. Set

$$A_{u,m}(n,b) = \frac{d(u\lambda^m n b^3)\alpha(u\lambda^m n b^3)\psi(u\lambda^{m+4} n b^3)}{3^{m/6}\sqrt{N(b)}}, \qquad |A_{u,m}(n,b)| \le 27.$$

The transformed term is therefore

$$C(k_0) \sum_{u \in \mathcal{O}^\times} \sum_{m \ge -4} \chi_{k_0}(u\lambda^{m+4})^3 \sum_{\substack{n,b \equiv 1 \ (3) \\ n \text{ squarefree}}} \frac{A_{u,m}(n,b)\chi_{k_0}(nb)^3}{3^{m/3}\sqrt{N(n)}\, N(b)}$$
$$\times \prod_{p \in \mathcal{A}_{\mathrm{fix}}} B_{p,j_p}(u\lambda^{m+4} n b^3) V_*^\sharp\Big(\frac{3^m N(n) N(b)^3 X}{N(c_*)^2 N(k_0)^2}\Big), \qquad |C(k_0)| \ll 1.$$

The representation is unique: away from λ, the prime exponents of ℓ are $3v_p(b)$ or $1 + 3v_p(b)$. Apart from the quadratic character and weight, the k_0-dependence is a bounded scalar.

Fix u, m and write $q = N(p)$ for an active prime $p \nmid k_0$. For $j_p = 4$, the local identity is

$$B_{p,4}(u\lambda^{m+4} n b^3) = -q^{-1/2} + q^{1/2}\mathbf{1}_{p \mid n} + q^{1/2}\mathbf{1}_{p \nmid n,\ p \mid b}.$$

To display the reindexing in (6.15), let $A^{(p)}(n,b)$ include all the other local factors and set

$$A^{(p,n)}(n,b) = \mathbf{1}_{p \nmid n} A^{(p)}(pn, b), \qquad A^{(p,b)}(n,b) = \mathbf{1}_{p \nmid n} A^{(p)}(n, pb).$$

Then the changes $n = pn'$ and $b = pb'$ give the exact identity

$$\begin{aligned} &\mathscr{S}_m[A^{(p)}(n,b) B_{p,4}(u\lambda^{m+4} n b^3), q^2 Y](k_0) \\ &\quad = -q^{-1/2}\mathscr{S}_m[A^{(p)}, q^2 Y](k_0) + \chi_{k_0}(p)^3 \mathscr{S}_m[A^{(p,n)}, qY](k_0) \\ &\qquad + q^{-1/2}\chi_{k_0}(p)^3 \mathscr{S}_m[A^{(p,b)}, Y/q](k_0). \end{aligned}$$

In the second term $p \nmid n'$ preserves squarefreeness; in the third, $p \nmid n$ is retained and b' is unrestricted at p. Both extracted phases have modulus one. The factors $q^{1/2}$ in the local identity cancel against $\sqrt{N(pn')}$ or leave $q^{-1/2}$ after division by $N(pb')$.

Let a track a bound $|A(n,b)| \le 27a$ for the coefficient in (6.15), and let Y be its scale. Before inserting the active primes, these are $a = 1$ and $Y = N(c_0)^2\mathcal{H}_0^2/X$. Each active prime contributes q^2 to the squared conductor norm. Including this factor, the updates are

$$(a^2, Y) \longmapsto \begin{cases} (a^2, q^2Y), & j_p \notin \{0,4\}, \\ (a^2/q, q^2Y), & j_p = 0, \\ (a^2/q, q^2Y),\ (a^2, qY),\ (a^2/q, Y/q), & j_p = 4. \end{cases}$$

Thus a^2 never increases. If $p \mid t$ and $p \notin S$, then j_p is odd, so a^2Y costs at most $N(p)^2$. If $p \mid g$ and $p \nmid t$, the cost is $N(p)$ for $j_p = 0, 4$ and $N(p)^2$ for $j_p = 2$; the latter case requires $v_p(g) \ge 2$. Hence

$$a^2Y \ll \frac{\mathcal{H}_0^2}{X} \prod_{\substack{p \mid t \\ p \notin S}} N(p)^2 \prod_{\substack{p \mid g \\ p \nmid t,\ p \notin S}} N(p)^{v_p(g)} \le \frac{\mathcal{H}_0^2 N(t)^2 N(g)}{X}.$$

Reindexing at distinct primes preserves the form and zero extensions. By Lemma 6.3, the transformed terms, units, and at most three choices per prime $p \in \mathcal{A}_{\mathrm{fix}}$ with $j_p = 4$ form an index set of divisor-bounded size in tg. Here and below, divisor-bounded in a means bounded by $C\tau_{\mathrm{div}}(a)^A$ for fixed constants C, A; in particular, this is $\ll_\varepsilon \mathrm{N}_{K/\mathbb{Q}}(a)^\varepsilon$ for every $\varepsilon > 0$. For one index ι, abbreviate $a = a_\iota$, $Y = Y_\iota$, and $A_m = A_{\iota,m}$. The local updates give $Y \ll D^{C_1}$ for some fixed $C_1 = C_1(C_0)$. Split b into $B \le N(b) < 2B$, and n by a smooth dyadic partition $V(N(n)/U)$, with $U, B \ge 1$ and uniformly bounded cutoffs. For this part of (6.15), write

$$\mathscr{S}_{m;U,B}(k_0) = 3^{-m/3} \sum_{B \le N(b) < 2B} \frac{\chi_{k_0}(b)^3}{N(b)} F_b(k_0),$$

$$F_b(k_0) = \sum_{n \text{ squarefree}} \frac{A_m(n,b)\chi_{k_0}(n)^3}{\sqrt{N(n)}} V(N(n)/U) V_*^\sharp\Big(\frac{3^m N(n) N(b)^3 \mathcal{H}_0^2}{Y N(k_0)^2}\Big).$$

Here n, b remain primary. Put $z_* = 3^m UB^3/Y$ and fix a decay exponent $A > 0$. With m, U, B, Y fixed, Lemma B.1, applied only in $N(n)/U$, gives the following representation in a real Mellin variable s:

$$F_b(k_0) = \int_{\mathbb{R}} c_{b,k_0}(s) G_{b,s}(k_0)\, ds,$$

$$G_{b,s}(k_0) = \sum_{\substack{n \text{ squarefree} \\ N(n) \asymp U}} \frac{A_m(n,b)\chi_{k_0}(n)^3}{\sqrt{N(n)}} (N(n)/U)^{\mathrm{i}s}, \tag{6.19}$$

$$|c_{b,k_0}(s)| \ll_{I,A} (1 + z_*)^{-A}(1 + |s|)^{-2}.$$

Indeed, the scale R in (B.2) is $3^m UN(b)^3\mathcal{H}_0^2/(YN(k_0)^2) \ge z_*$. Lemma 6.4 supplies the required derivative bounds, independently of b, k_0, U, B, m. Since $\sum_{N(n) \asymp U} N(n)^{-1} \ll 1$, Lemma 6.5 gives, for any $\sigma > 0$,

$$\sideset{}{^*}\sum_{N(k_0) \le \mathcal{H}_0} |G_{b,s}(k_0)|^2 \ll_\sigma a^2(\mathcal{H}_0 U)^\sigma(\mathcal{H}_0 + U).$$

Apply weighted Cauchy–Schwarz to the Mellin integral, as recorded in (B.6), with the common majorant in (6.19), then weighted Cauchy–Schwarz in b, using $\sum_{B \le N(b) < 2B} N(b)^{-1} \ll 1$:

$$\begin{aligned} \sideset{}{^*}\sum_{N(k_0) \le \mathcal{H}_0} |\mathscr{S}_{m;U,B}(k_0)|^2 &\le 3^{-2m/3}\Big(\sum_{B \le N(b) < 2B} \frac{1}{N(b)}\Big) \sum_{B \le N(b) < 2B} \frac{1}{N(b)} \sideset{}{^*}\sum_{N(k_0) \le \mathcal{H}_0} |F_b(k_0)|^2 \\ &\ll_{I,A,\sigma} a^2 3^{-2m/3}(\mathcal{H}_0 U)^\sigma(\mathcal{H}_0 + U)(1 + z_*)^{-2A}. \end{aligned} \tag{6.20}$$

Since $m \ge -4$, we have $U \le 81 Y z_*$ and hence $\mathcal{H}_0 U \ll D^{C_0+C_1}(1+z_*)$. Choose $0 < \sigma \le 1$ small in terms of $\varepsilon_1 > 0, C_0$ and take $A = 3$. Taking square roots in (6.20) gives

$$\Big(\sideset{}{^*}\sum_{N(k_0)\le \mathcal{H}_0} |\mathscr{S}_{m;U,B}(k_0)|^2 \Big)^{1/2} \ll a D^{\varepsilon_1} 3^{-m/3} (\sqrt{\mathcal{H}_0} + \sqrt{U})(1 + 3^m U B^3/Y)^{-2}.$$

The infinite dyadic sums converge: for dyadic $U, B \ge 1$ and every $Z > 0$,

$$\sum_{U,B \text{ dyadic}} (1 + UB^3/Z)^{-2} \ll \log^2(2+Z), \qquad \sum_{U,B \text{ dyadic}} \sqrt{U}(1 + UB^3/Z)^{-2} \ll \sqrt{Z}.$$

These estimates make the sum of $\ell^2(k_0)$ norms finite. The triangle inequality and the geometric sum over $m \ge -4$ therefore give

$$\Big(\sideset{}{^*}\sum_{N(k_0)\le \mathcal{H}_0} \Big| \sum_{m \ge -4} c_{\iota,m}(k_0) \mathscr{S}_m[A_m, Y](k_0) \Big|^2 \Big)^{1/2} \ll a D^{\varepsilon_1} \big(\sqrt{\mathcal{H}_0} \log^2(2+Y) + \sqrt{Y} \big).$$

Combine the divisor-bounded choices of ι and use (6.16) to obtain

$$\sideset{}{^*}\sum_{N(k_0)\le \mathcal{H}_0} |T(X; u_0 t k_0, g)|^2 \preccurlyeq \max_{\iota} a_\iota^2 (\mathcal{H}_0 + Y_\iota) \ll \mathcal{H}_0 + \frac{\mathcal{H}_0^2 N(t)^2 N(g)}{X} \le \mathcal{H} + \frac{\mathcal{H}^2 N(g)}{X}.$$

Sum the fixed ray classes and the divisor-bounded choices $t \mid \mathrm{rad}(g \prod_{\mathfrak{p} \in S} \mathfrak{p})$, choosing ε_1 and the other small-power losses in terms of ε. The weight estimates above require a fixed number $J = J(\varepsilon, C_0)$ of derivatives. Homogeneity restores $\|W\|_{C^J(I)}^2$ and proves (6.14). □

Proof of Proposition 5.2. Write uniquely $k = u_0 s v^2$, with s squarefree and s, v among the chosen ideal generators. No condition $(s, v) = 1$ is imposed. Since $8 \equiv 2 \pmod 6$, including the zero extensions,

$$\chi_n(u_0 s v^2) \chi_n(f)^4 = \chi_n(u_0 s) \chi_n(f v^2)^4, \qquad T(X; u_0 s v^2, f) = T(X; u_0 s, f v^2).$$

Apply Lemma 6.6 with row bound $\mathcal{H}/N(v)^2$ and auxiliary twist $g = f v^2$. Here $N(g) \le \mathcal{H} N(f) \le D^{2C_0}$, so use that lemma with $2C_0$. Summing its bounds gives

$$\begin{aligned} \sum_{0 < N(k) \le \mathcal{H}} |T(X; k, f)|^2 &= \sum_{u_0 \in \mathcal{O}^\times} \sum_{N(v)^2 \le \mathcal{H}} \sum_{\substack{N(s) \le \mathcal{H}/N(v)^2 \\ s \text{ squarefree}}} |T(X; u_0 s, f v^2)|^2 \\ &\ll D^{\varepsilon} \|W\|_{C^J(I)}^2 \Big(\mathcal{H} + \frac{\mathcal{H}^2 N(f)}{X} \Big) \sum_v \frac{1}{N(v)^2}. \end{aligned}$$

The last sum is $\zeta_K(2) < \infty$, where ζ_K is the Dedekind zeta function of K (Input E). This proves (5.5). □

7 Proof of the transfer proposition

We prove Proposition 5.4 for $\mathcal{A}(W)$ defined in (5.10). Throughout this section, $\mathcal{H}, L, F, \Sigma, \xi$ satisfy the hypotheses of that proposition. We retain the fixed weight Φ and the support bound C_Φ chosen before (5.10). The enlargement by $\Sigma/(LF) \ge 1$ in the intermediate mean square makes the second Poisson summation return to (4.9) with row range at most $\mathcal{H}L/(\Sigma F)$. Lemma B.2 separates the weights, and Lemma 4.4 removes the remaining exclusion.

7.1 First application of Poisson summation

Write $N = \mathrm{N}_{K/\mathbb{Q}}$, $I = [u, v]$, and $I_* = [u/2, 2v]$. All ideal indices below are prime to S and represented by their primary generators; a star additionally requires squarefreeness. The variables k, h, y range over $\mathcal{O}$. For squarefree C, t with $(C, t) = 1$, and for $d \mid C$, put

$$\ell = \frac{L}{N(C)N(t)}, \qquad w_{C,d} = \frac{\mathcal{H}N(C)}{N(d)L^2F}, \qquad Y_{C,d} = c_I \frac{\Sigma LFN(d)}{\mathcal{H}N(C)^2},$$

where $c_I \geq \max(1, 4C_\Phi v^2)$ is fixed. Retain only $\ell \geq 1/(2v)$. For $U \in C_c^\infty(I_*)$ and a character ξ_1 of the fixed ray class group, define

$$\begin{aligned} p_y(n) &= \mu(n)\xi_1(n)\mathbf{1}_{(n,t)=1}\overline{\chi_n(y)}\chi_n(C)^4\chi_n(d), \\ P_{C,d,t}(y;U) &= \sum_n{}^{*} p_y(n)U(N(n)/\ell). \end{aligned} \tag{7.1}$$

The notation p_y suppresses its dependence on C, d, t, ξ_1. The nonnegative form required in the second Poisson calculation is

$$\mathcal{Q}_{\xi_1}(U) = \sum_{\substack{C,t \text{ squarefree} \\ (Ct,S)=1,(C,t)=1 \\ N(C)N(t)\leq 2vL}} \sum_{d\mid C} w_{C,d} \sum_y \Phi(N(y)/Y_{C,d})|P_{C,d,t}(y;U)|^2. \tag{7.2}$$

The outer domain has no additional restriction coming from the support of $\widehat{\Phi}$. Here $Y_{C,d} > 0$ may be smaller than 1; the outer sums are finite, and the y-sum converges absolutely.

Lemma 7.1. *Under the hypotheses of Proposition 5.4, fix an integer $j \geq 0$ and $\varepsilon_0 > 0$. If $M \geq 0$ satisfies $\mathcal{Q}_{\xi_1}(U) \leq M\|U\|^2_{C^j(I_*)}$ for every $U \in C_c^\infty(I_*)$ and every ξ_1, then*

$$\mathcal{A}(W) \ll D^{\varepsilon_0}(\Sigma + M)\|W\|^2_{C^{2j+4}(I)}.$$

Proof. Expand the square defining $\mathcal{A}(W)$, and write $n_i = Cu_i$, where $C = (n_1, n_2)$ and $(u_1, u_2) = (u_1u_2, C) = 1$. The row character is $\chi_{u_1}\overline{\chi_{u_2}}$, primitive modulo u_1u_2, with the extra restriction $(k, C) = 1$. Lemma 4.2 introduces $d \mid C$ and a frequency h. Let Z denote the zero-frequency contribution. It requires $u_1 = u_2 = 1$, so

$$|Z| \ll \frac{\mathcal{H}}{LF} \sum_{F\leq N(f)<2F}{}^{*} \sum_C{}^{*} |W(N(C)/L)|^2 \ll_I \mathcal{H}\|W\|^2_\infty.$$

For the other frequencies, the Chinese remainder theorem, (4.4), and (4.7) give the paired identity

$$a_\xi(u_1)\overline{a_\xi(u_2)}\gamma(\chi_{u_1}\overline{\chi_{u_2}}) = \mu(u_1)\mu(u_2)(\xi G)(u_1u_2^{-1}). \tag{7.3}$$

Expand $(\xi G)(z) = \sum_{\xi_1} \ell_{\xi_1}\xi_1(z)$ on the fixed ray class group. Insert $\mathbf{1}_{(u_1,u_2)=1} = \sum_{t\mid u_1,\ t\mid u_2} \mu(t)$ and put $u_i = tx_i$. The inverse is $x_i = u_i/t$; squarefreeness imposes $(x_i, t) = 1$, but no condition $(x_1, x_2) = 1$ remains. The factors at C, t cancel against their conjugates, leaving $\mu(d)\mu(t)$ and the restrictions $(f, Ct) = (h, t) = 1$. Thus

$$\begin{aligned} \mathcal{A}(W) - Z = \sum_{\xi_1} \ell_{\xi_1} &\sum_{\substack{C,t \text{ squarefree} \\ (Ct,S)=1,(C,t)=1 \\ N(C)N(t)\leq 2vL}} \sum_{d\mid C} \mu(d)\mu(t)w_{C,d} \\ \times \sum_{\substack{F\leq N(f)<2F \\ (f,Ct)=1}}{}^{*} &\sum_{\substack{h\neq 0,(h,t)=1 \\ N(h)\leq C_\Phi v^2L^2N(d)/(\mathcal{H}N(C)^2)}} \sum_{x_1,x_2}{}^{*} p_{hf^2}(x_1)\overline{p_{hf^2}(x_2)}\mathscr{K}_h(N(x_1)/\ell, N(x_2)/\ell), \end{aligned}$$

where, for each integer $j_0 \ge 0$,

$$\mathscr{K}_h(z_1,z_2) = (z_1z_2)^{-1/2}W(z_1)\overline{W(z_2)}\widehat{\Phi}\Big(\frac{\mathcal{H}N(h)N(C)^2}{L^2N(d)z_1z_2}\Big),$$
$$\sup_{C,d,h} \|\mathscr{K}_h\|_{C^{j_0}(I^2)} \ll_{I,\Phi,j_0} \|W\|^2_{C^{j_0}(I)}.$$

Here $\chi_{x_i}(C)^4$ supplies $(x_i, C) = 1$; the other original zeros are supplied by $p_{hf^2}(x_i)$. The displayed frequency range follows from the support of $\widehat{\Phi}$.

For fixed C, d, t, the map $(f,h) \mapsto y = hf^2$ has divisor-bounded multiplicity in y. Since $\Sigma/(LF) \ge 1$, its nonzero image satisfies $N(y) \le 4C_\Phi v^2L^2F^2N(d)/(\mathcal{H}N(C)^2) \le Y_{C,d}$. Since $\Phi \ge 1$ on $[0,1]$ and is nonnegative,

$$\sum_{C,t}\sum_{d\mid C} w_{C,d}\sum_{f,h}|P_{C,d,t}(hf^2;U)|^2 \preccurlyeq \mathcal{Q}_{\xi_1}(U) \le M\|U\|^2_{C^j(I_*)},$$

with the preceding ranges on the left. Apply Lemma B.2 to the full displayed expression for $\mathcal{A}(W) - Z$. The kernel bound gives

$$|\mathcal{A}(W) - Z| \ll D^{\varepsilon_0/2}M\|W\|^2_{C^{2j+4}(I)}.$$

Together with $\mathcal{H} \le \Sigma$, this proves the result after allocating the divisor losses within ε_0. □

7.2 Second application of Poisson summation

Let C, t be coprime squarefree primary elements prime to S, and let $d \mid C$. Fix a character ξ_1 of the ray class group. For arbitrary scales $\ell, Y > 0$ and $U \in C_c^\infty(I_*)$, with $I_* = [a,b] \subset (0,\infty)$, put

$$\begin{aligned} P(y) &= \sideset{}{^*}\sum_{(n,S)=1} \mu(n)\xi_1(n)\mathbf{1}_{(n,t)=1}\overline{\chi_n(y)}\chi_n(C)^4\chi_n(d)U(N(n)/\ell), \\ \mathcal{M} &= \sum_{y\in\mathcal{O}} \Phi(N(y)/Y)|P(y)|^2. \end{aligned} \tag{7.4}$$

Here $P(y)$ abbreviates $P_{C,d,t}(y;U)$ from (7.1), with the scale ℓ now arbitrary.

The second Poisson summation turns the Möbius coefficients back into cubic Gauss-sum coefficients. To state the identity, put $U_0(x) = x^{-1/2}U(x)$ and expand

$$\xi_1(z)G(z^{-1}) = \sum_{\xi'} c_{\xi'}\xi'(z)$$

on the fixed ray class group.

On the transformed side, g is the common divisor of the original columns, $e \mid g$ comes from the row exclusion, and w removes the remaining coprimality condition. The index h is the nonzero Fourier frequency. Let g, w range over squarefree primary elements prime to S, e over divisors of g, and h over $\mathcal{O}\setminus\{0\}$, subject to

$$(g, Ct) = (w, gCth) = 1, \qquad N(gw) \le b\ell, \qquad N(h) \le \frac{C_\Phi b^2\ell^2N(e)}{YN(g)^2}.$$

These conditions ensure that tg/e and Cew are coprime and squarefree. For each such choice and each character ξ', the new column scale, coefficient, and coupled smooth weight are

$$\begin{aligned} X' &= \frac{\ell}{N(gw)}, \\ a^\sharp(n) &= a_{\xi'}(n)\mathbf{1}_{(n,tg/e)=1}\chi_n(deh)\chi_n(Cew)^4, \\ \mathscr{K}(x_1,x_2) &= U_0(x_1)\overline{U_0(x_2)}\widehat{\Phi}\Big(\frac{YN(h)N(g)^2}{N(e)\ell^2x_1x_2}\Big). \end{aligned} \tag{7.5}$$

Here $\mathscr{K}$ depends on U, Y, ℓ, g, e, h. A star on a sum over n_1, n_2 restricts each index separately to squarefree primary elements. Define also

$$Z = Y\widehat{\Phi}(0) \sideset{}{^*}\sum_{\substack{(g,Ct)=1\\(g,S)=1}} |U(N(g)/\ell)|^2 \prod_{p\mid g}\Big(1-\frac{1}{N(p)}\Big).$$

Lemma 7.2 (Second Poisson identity)**.** *With the notation and index ranges above,*

$$\mathcal{M} = Z + \sum_{g,e,w,h,\xi'} \frac{Y\mu(e)\mu(w)c_{\xi'}N(g)}{N(e)\ell} \sideset{}{^*}\sum_{(n_1n_2,S)=1} a^\sharp(n_1)\overline{a^\sharp(n_2)}\mathscr{K}\Big(\frac{N(n_1)}{X'},\frac{N(n_2)}{X'}\Big). \tag{7.6}$$

The zero-frequency term satisfies

$$|Z| \ll_{I_*,\Phi} Y\ell\|U\|_\infty^2.$$

Proof. Expand the square, put $g=(n_1,n_2)$ and $n_i = gz_i$. The nonzero terms satisfy

$$(z_1,z_2) = (z_1z_2, gCt) = (g,Ct) = 1, \qquad \overline{\chi_{n_1}(y)}\chi_{n_2}(y) = \mathbf{1}_{(y,g)=1}\overline{\chi_{z_1}(y)}\chi_{z_2}(y).$$

Apply Lemma 4.2 to this last character, with exclusion g and divisor $e \mid g$. Its zero frequency occurs exactly when $z_1 = z_2 = 1$ and gives the displayed Z; counting g gives its bound. For the nonzero frequencies, (4.5)–(4.7) and the Chinese remainder theorem give

$$\mu(z_1)\mu(z_2)\xi_1(z_1)\overline{\xi_1(z_2)}\gamma(\overline{\chi_{z_1}}\chi_{z_2}) = \sum_{\xi'} c_{\xi'}a_{\xi'}(z_1)\overline{a_{\xi'}(z_2)}.$$

Use also

$$\chi_z(dh)\overline{\chi_z(e)}\chi_z(C)^4 = \chi_z(deh)\chi_z(Ce)^4,$$
$$\frac{U(N(gz_1)/\ell)\overline{U(N(gz_2)/\ell)}}{\sqrt{N(z_1)N(z_2)}} = \frac{N(g)}{\ell}U_0(N(gz_1)/\ell)\overline{U_0(N(gz_2)/\ell)}.$$

Thus the full nonzero contribution is

$$\begin{aligned}\mathcal{M} - Z = \sum_{\xi'} c_{\xi'} \sideset{}{^*}\sum_{\substack{g\\(g,Ct)=1}} \sum_{e\mid g} \frac{Y\mu(e)N(g)}{N(e)\ell} \sum_{h\neq 0} \\ \times \sideset{}{^*}\sum_{\substack{z_1,z_2\\(z_1,z_2)=1\\(z_1z_2,gCt)=1}} a_{\xi'}(z_1)\overline{a_{\xi'}(z_2)}\chi_{z_1}(deh)\overline{\chi_{z_2}(deh)}\chi_{z_1}(Ce)^4\overline{\chi_{z_2}(Ce)^4} \\ \times U_0\Big(\frac{N(gz_1)}{\ell}\Big)\overline{U_0\Big(\frac{N(gz_2)}{\ell}\Big)}\widehat{\Phi}\Big(\frac{YN(h)}{N(e)N(z_1)N(z_2)}\Big).\end{aligned}$$

All starred ideal indices are prime to S. Insert

$$\mathbf{1}_{(z_1,z_2)=1} = \sum_{w\mid z_1,\ w\mid z_2} \mu(w), \qquad z_i = wn_i.$$

Here $(w, gCt) = 1$. Equation (4.6) and the zero-extended characters give

$$a_{\xi'}(wn) = a_{\xi'}(w)a_{\xi'}(n)\chi_n(w)^4 \quad ((w,n)=1),$$
$$|a_{\xi'}(w)\chi_w(deh)\chi_w(Ce)^4|^2 = \mathbf{1}_{(w,h)=1},$$
$$\mathbf{1}_{(n,gCtw)=1}\chi_n(deh)\chi_n(Cew)^4 = \mathbf{1}_{(n,tg/e)=1}\chi_n(deh)\chi_n(Cew)^4.$$

Consequently the preceding full sum becomes

$$\mathcal{M}-Z=\sum_{\xi'}c_{\xi'}\sideset{}{^*}\sum_{\substack{g,w\\(g,Ct)=(w,gCt)=1}}\sum_{e\mid g}\frac{Y\mu(e)\mu(w)N(g)}{N(e)\ell}\sum_{\substack{h\neq0\\(h,w)=1}}$$
$$\times\sideset{}{^*}\sum_{(n_1n_2,S)=1}a^\sharp(n_1)\overline{a^\sharp(n_2)}U_0\Big(\frac{N(gwn_1)}{\ell}\Big)\overline{U_0\Big(\frac{N(gwn_2)}{\ell}\Big)}\widehat{\Phi}\Big(\frac{YN(h)}{N(e)N(w)^2N(n_1)N(n_2)}\Big).$$

There is no remaining condition $(n_1,n_2)=1$. The column support gives $N(gw)\le b\ell$ and $N(z_1z_2)\le b^2\ell^2/N(g)^2$; the support of $\widehat{\Phi}$ therefore imposes the stated bound on $N(h)$. Substituting (7.5) gives (7.6), with only finitely many terms. □

Lemma 7.3. *Return to $I=[u,v]$ and $I_*=[u/2,2v]$. For an integer $m\ge0$, let S_m be the supremum in (5.11). Then, for every $\varepsilon_0>0$,*

$$\mathcal{Q}_{\xi_1}(U)\ll D^{\varepsilon_0}\Sigma(1+S_m)\|U\|^2_{C^{2m+4}(I_*)}\qquad(U\in C_c^\infty(I_*)).$$

The test functions in S_m are supported in $I'=[u/16,4v]$ and have $C^m(I')$ norm at most one; an empty supremum is zero.

Proof. Write $N=\mathrm{N}_{K/\mathbb{Q}}$. Apply Lemma 7.2 with input length ℓ, row scale $Y_{C,d}$, and the same C,d,t, using the character ξ_1. Here h is the new Fourier variable for the sum over y. Its nonzero output has squarefree g,w, $e\mid g$, and

$$(g,Ct)=(w,gCth)=1,\qquad r=tg/e,\quad f'=Cew,\quad k'=deh.$$

Thus r,f' are coprime and squarefree, and the coefficient $a^\sharp$ from (7.5) becomes

$$a^\sharp(n)=a_{\xi'}(n)\mathbf{1}_{(n,r)=1}\chi_n(k')\chi_n(f')^4.$$

Write X_0' for the individual column scale X' in (7.5); below, X' will denote a common dyadic scale. The length, coefficient, and Fourier parameter simplify to

$$X_0'=\frac{\ell}{N(g)N(w)}=\frac{L}{N(r)N(f')},$$
$$w_{C,d}\frac{Y_{C,d}N(g)}{N(e)\ell}=\frac{c_I\Sigma N(r)}{L^2},$$
$$\frac{Y_{C,d}N(h)N(g)^2}{N(e)\ell^2}=\frac{c_I\Sigma FN(k')N(r)^2}{\mathcal{H}L}.$$

Put $U_0(x)=x^{-1/2}U(x)$. The coupled kernel is therefore

$$\mathscr{K}_{r,f',k'}(x_1,x_2)=U_0(x_1)\overline{U_0(x_2)}\widehat{\Phi}\Big(\frac{c_I\Sigma FN(k')N(r)^2}{\mathcal{H}Lx_1x_2}\Big),$$

with both columns evaluated at $x_j=N(n_j)/X_0'$. In particular its nonzero support requires

$$N(r)N(f')\le2vL,\qquad0<N(k')\le\frac{4C_\Phi v^2}{c_I}\frac{\mathcal{H}L}{\Sigma FN(r)^2}\le\frac{\mathcal{H}L}{\Sigma FN(r)^2},$$

since $C_\Phi(2v)^2/c_I\le1$.

For fixed r,f',k', all preimages are obtained by choosing

$$t\mid r,\qquad f'=Cew,\qquad e\mid k',\qquad(w,k')=1,\qquad d\mid(C,k'),$$

and setting $g = e(r/t)$, $h = k'/(de)$. The factorization $f' = Cew$ is disjoint and squarefree. All these preimages have the same kernel: $Ctgw = rf'$, and the Fourier parameter displayed above depends only on r, k'. Thus the support restrictions discard only zero kernels. The choices $t \mid r$ contribute $\tau_{\mathrm{div}}(r)$. At a prime $p \mid f'$, the choices $p \mid C$, $p \mid e$, and $p \mid w$, respectively, contribute

$$\big(1 + \mathbf{1}_{p|k'}\big) - \mathbf{1}_{p|k'} - \mathbf{1}_{p\nmid k'} = \mathbf{1}_{p|k'}$$

to the sum of $\mu(e)\mu(w)$ over the preimages; the two choices inside the first term are $p \nmid d$ and $p \mid d$. Let $\mathcal{Z}$ be the total zero-frequency contribution: the sum of the terms Z from Lemma 7.2, with the outer weights $w_{C,d}$ and ranges in (7.2). Consequently, regrouping before taking absolute values gives

$$\mathcal{Q}_{\xi_1}(U) - \mathcal{Z} = \frac{c_I \Sigma}{L^2} \sum_{\xi'} c_{\xi'} \sum_{\substack{r, f' \text{ squarefree} \\ (r,f')=1}} N(r)\tau_{\mathrm{div}}(r) \sum_{\substack{k' \neq 0 \\ f' \mid k'}} \times \sideset{}{^*}\sum_{n_1,n_2} a^\sharp(n_1)\overline{a^\sharp(n_2)}\mathscr{K}_{r,f',k'}\Big(\frac{N(n_1)}{X_0'}, \frac{N(n_2)}{X_0'}\Big).$$

Fix $R \le N(r) < 2R$, $F' \le N(f') < 2F'$ dyadically, and put

$$X' = \frac{L}{RF'}, \qquad \Sigma' = X'F' = \frac{L}{R}, \qquad \mathcal{H}' = \frac{\mathcal{H}L}{\Sigma F R^2}.$$

Retain only those r for which a nonzero term occurs in these ranges. Since $f' \mid k'$, each such r satisfies

$$F' \le N(f') \le N(k') \le \frac{\mathcal{H}L}{\Sigma F N(r)^2}.$$

In particular $\mathcal{H}' \ge 1$. For every $j \mid r$, using $\mathcal{H} \le \Sigma$, we also have

$$\frac{X'}{N(j)} \ge \frac{\Sigma F N(r)^2}{\mathcal{H} R N(j)} \ge \frac{F\Sigma}{\mathcal{H}} \ge 1. \tag{7.7}$$

Moreover $1 \le X'/X_0' < 4$. The rescaled kernels

$$\mathcal{K}_{r,f',k'}(x_1, x_2) = \mathscr{K}_{r,f',k'}\Big(\frac{X'}{X_0'}x_1, \frac{X'}{X_0'}x_2\Big)$$

have support in $[u/8, 2v]^2 \subset \mathrm{int}(I'^2)$ and satisfy

$$\sup_{r,f',k'} \|\mathcal{K}_{r,f',k'}\|_{C^{2m+4}(I'^2)} \ll_{I_*,m,\Phi} \|U\|^2_{C^{2m+4}(I_*)},$$

where the supremum is over the retained indices in the dyadic block. This follows from the Schwartz bounds for $\widehat{\Phi}$. The common parameters satisfy exactly

$$\frac{\mathcal{H}'}{\Sigma'} = \frac{\mathcal{H}}{\Sigma F R} \le \frac{\mathcal{H}}{\Sigma}, \qquad \Sigma' \le L, \qquad \mathcal{H}' \le \frac{\mathcal{H}L}{\Sigma F}.$$

Fix $\delta > 0$, to be chosen in terms of ε_0 at the end. For each retained r, enlarge the f', k' ranges by positivity, keeping this set of r fixed. Lemma 4.4 then gives, for $V \in C_c^\infty(I')$,

$$\sideset{}{^*}\sum_{F' \le N(f') < 2F'} \sum_{0 < N(k') \le \mathcal{H}'} \Big|\sideset{}{^*}\sum_n a^\sharp(n) V(N(n)/X')\Big|^2 = \Sigma' \mathcal{E}_{(r)}(\mathcal{H}', X', F'; \xi', V) \ll_\delta D^\delta (\Sigma')^2 S_m \|V\|^2_{C^m(I')}.$$

Indeed exclusion removal replaces (X', F') by $(X'/N(j), F'N(j))$, $j \mid r$, preserving $\mathcal{H}'$ and Σ'. Equation (7.7) gives $X'/N(j) \ge 1$, and $F'N(j) \ge 1$ as well. Thus every resulting mean square lies in the defining supremum.

There are $O(R)$ ideals r in the dyadic range. Apply Lemma B.2 with row index (r, f', k'), the preceding mean-square bound, and the uniform kernel bound. The absolute contribution of this dyadic block is at most

$$D^{2\delta}\frac{\Sigma R}{L^2}\,R(\Sigma')^2 S_m\|U\|^2_{C^{2m+4}(I_*)} = D^{2\delta}\Sigma S_m\|U\|^2_{C^{2m+4}(I_*)},$$

because $(\Sigma R/L^2)R(L/R)^2 = \Sigma$.

Finally, the zero-frequency bound in Lemma 7.2 gives

$$\begin{aligned}|\mathcal{Z}| &\ll_{I_*,\Phi} \|U\|_\infty^2 \sum_{N(Ct)\le 2vL}\sum_{d\mid C} w_{C,d}Y_{C,d}\ell \\ &\ll_{I_*} \Sigma\|U\|_\infty^2 \sum_C \frac{\tau_{\mathrm{div}}(C)}{N(C)^2}\sum_{N(t)\le 2vL}\frac{1}{N(t)} \ll_{I_*,\delta} D^\delta\Sigma\|U\|_\infty^2.\end{aligned}$$

This uses only $Y_{C,d} > 0$, not $Y_{C,d} \ge 1$. Sum the $O_{I_*,C_0}((\log(2D))^2)$ dyadic blocks and the fixed finite character set, and take $\delta = \varepsilon_0/4$. The stated bound follows, with implied constant depending only on $I_*, m, C_0, \varepsilon_0$, the fixed cutoffs and arithmetic data. □

Proof of Proposition 5.4. Use S_m as in Lemma 7.3, with m as in the proposition. Lemma 7.3, with loss $D^{\varepsilon/4}$, supplies the hypothesis of Lemma 7.1 with $j = 2m+4$ and $M \ll D^{\varepsilon/4}\Sigma(1+S_m)$. Applying that lemma with the same loss gives

$$\mathcal{A}(W) \ll D^{\varepsilon/2}\Sigma(1+S_m)\|W\|^2_{C^{4m+12}(I)},$$

which proves (5.11). □

A Arithmetic identities and theta calculations

A.1 Reciprocity and Gauss sums

Proof of Lemma 4.1. Proof of (4.2). For squarefree n, set $G(n) = \overline{\chi_n(4)}\gamma_3(n)$. Its dependence on a fixed ray class will be verified below. Fix a primary prime $p \equiv 1 \pmod 3$ outside S. For multiplicative characters A, B of $(\mathcal{O}/(p))^\times$, extended by zero at 0, write

$$J(A,B) = \sum_{x \bmod p} A(x)B(1-x).$$

For each $y \in \mathcal{O}/(p)$, the substitution $u = 2x-1$ gives

$$\#\{x \bmod p : 4x(1-x) = y\} = \#\{u \bmod p : u^2 = 1-y\} = 1 + \chi_p^3(1-y),$$

since $p \nmid 2$ and χ_p^3 is the quadratic character, extended by zero at 0. Hence

$$\chi_p(4)J(\chi_p,\chi_p) = \sum_{x \bmod p}\chi_p\big(4x(1-x)\big) = \sum_{y \bmod p}\chi_p(y)\big(1+\chi_p^3(1-y)\big) = J(\chi_p,\chi_p^3).$$

The Gauss–Jacobi identity (cf. [14, (3.18)]) gives

$$\frac{\gamma_1(p)\gamma_3(p)}{\gamma_4(p)} = \frac{J(\chi_p,\chi_p^3)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(p)}} = \chi_p(4)\frac{J(\chi_p,\chi_p)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(p)}} = \chi_p(4)\frac{\gamma_1(p)^2}{\gamma_2(p)}.$$

For $j \not\equiv 0 \pmod 6$, the normalized Gauss sums satisfy

$$\gamma_j(p)\gamma_{-j}(p) = \chi_p(-1)^j.$$

Taking $j = 2$ gives $\gamma_2(p)\gamma_4(p) = \chi_p^2(-1) = 1$, since χ_p^2 is cubic. Combining this with the Gauss–Jacobi relation above gives, after cancelling $\gamma_1(p)$ and rearranging,

$$\gamma_1(p)\gamma_2(p) = \overline{\chi_p(4)}\gamma_3(p)\gamma_2(p)^3.$$

To finish the prime case of (4.2), it remains to evaluate $\gamma_2(p)^3$. We do so by computing $J(\chi_p^2, \chi_p^2)$. Put $q = \mathrm{N}_{K/\mathbb{Q}}(p)$. We have $J(\chi_p^2, \chi_p^2) \in \mathcal{O}$ and $\mathrm{N}_{K/\mathbb{Q}}\big(J(\chi_p^2, \chi_p^2)\big) = q$. For $0 \le \ell \le (q-1)/3$, the exponent $(q-1)/3 + \ell$ lies strictly between 0 and $q-1$, so $\sum_{x \bmod p} x^{(q-1)/3+\ell} = 0$ in $\mathcal{O}/(p)$. Reducing modulo p and expanding the binomial therefore gives

$$\begin{aligned} J(\chi_p^2, \chi_p^2) &\equiv \sum_{x \bmod p} x^{(q-1)/3}(1-x)^{(q-1)/3} \\ &= \sum_{\ell=0}^{(q-1)/3} (-1)^\ell \binom{(q-1)/3}{\ell} \sum_{x \bmod p} x^{(q-1)/3+\ell} = 0 \pmod{p}. \end{aligned}$$

Thus $J(\chi_p^2, \chi_p^2)/p \in \mathcal{O}$ has norm one, so $J(\chi_p^2, \chi_p^2)$ is a unit multiple of p. To determine the unit, write $\chi_p^2(x(1-x)) = \omega^{j_x}$ with $j_x \in \{0, 1, 2\}$ for $x \in \mathcal{O}/(p) \setminus \{0, 1\}$. Then

$$\prod_{x \neq 0,1} x(1-x) = 1 \quad \Longrightarrow \quad \sum_{x \neq 0,1} j_x \equiv 0 \pmod{3}.$$

Since $(\omega - 1)^2$ generates (3), expansion modulo this ideal gives

$$J(\chi_p^2, \chi_p^2) = \sum_{x \neq 0,1} \omega^{j_x} \equiv (q-2) + (\omega - 1) \sum_{x \neq 0,1} j_x \equiv -1 \pmod{3}.$$

Together with $p \equiv 1 \pmod 3$, this fixes the unit:

$$J(\chi_p^2, \chi_p^2) = -p.$$

Consequently

$$\gamma_2(p)^3 = \frac{\gamma_2(p)^2}{\gamma_4(p)} = \frac{J(\chi_p^2, \chi_p^2)}{\sqrt{\mathrm{N}_{K/\mathbb{Q}}(p)}} = -\alpha(p).$$

The Chinese remainder theorem extends these prime-modulus identities to (4.2) for squarefree n, with one minus sign for each prime factor accounting for $\mu(n)$.

Proof of (4.6). For coprime squarefree primary a, b prime to S, the Chinese remainder theorem gives

$$\gamma_2(ab) = \gamma_2(a)\gamma_2(b)\chi_a(b)^2\chi_b(a)^2,$$

and cubic reciprocity gives $\chi_a(b)^2 = \chi_b(a)^2$. Multiplying by $\overline{\alpha(ab)}\xi(ab)$ therefore proves (4.6).

Proof of (4.1). We now prove (4.1) and the formula for $G(n)$ in (4.3), including their dependence on a fixed ray class. We begin by evaluating the normalized quadratic Gauss sum. For $c \in \mathcal{O}$ coprime to 2, we will show that

$$\Gamma_{\mathrm{quad}}(c) := |c|^{-1} \sum_{x \bmod c} e(x^2/c) = \frac{1}{2} \sum_{y \bmod 2\mathcal{O}} e(-cy^2/4). \tag{A.1}$$

To justify (A.1), apply Poisson summation over $z \in \mathcal{O}$ to $e(z^2/c)e^{-\pi\eta|z|^2}$, with $\eta > 0$. The Gaussian makes the sum convergent. The Fourier transform of the product is

$$\frac{|c|}{2\sqrt{r_\eta}} e^{-\pi\eta \mathrm{N}_{K/\mathbb{Q}}(c)|y|^2/(4r_\eta)} e(-cy^2/(4r_\eta)), \qquad r_\eta = 1 + 3\mathrm{N}_{K/\mathbb{Q}}(c)\eta^2/16.$$

Divide both sides by the Gaussian mass

$$\sum_{z\in\mathcal{O}} e^{-\pi\eta|z|^2} \asymp \eta^{-1}.$$

On the Fourier side, replacing r_η by 1 has total error

$$O_c\Big(\eta^2 \sum_{y\in\mathcal{O}} |y|^2 e^{-C_c\eta|y|^2}\Big) = O_c(1),$$

where $C_c > 0$ is fixed. The normalized error is therefore $O_c(\eta) \to 0$. Grouping the original sum modulo c and the Fourier sum modulo $2\mathcal{O}$, then letting $\eta \downarrow 0$, gives (A.1).

For $c = a + b\omega$, formula (A.1) becomes

$$\Gamma_{\mathrm{quad}}(a+b\omega) = \frac{1 + \mathrm{i}^{-b} + \mathrm{i}^{a} + \mathrm{i}^{b-a}}{2}.$$

Thus $\Gamma_{\mathrm{quad}}(c)$ depends only on $c \bmod 4\mathcal{O}$ and is invariant under multiplication by a square in $(\mathcal{O}/4\mathcal{O})^\times$. Representatives for the four square classes and their values are

c	1	-1	λ	$-\lambda$
$\Gamma_{\mathrm{quad}}(c)$	1	1	i	$-\mathrm{i}$

Consequently the function

$$\mathcal{R}(c_1, c_2) := \frac{\Gamma_{\mathrm{quad}}(c_1c_2)}{\Gamma_{\mathrm{quad}}(c_1)\Gamma_{\mathrm{quad}}(c_2)}$$

on these square classes satisfies

$$\mathcal{R}((-1)^e\lambda^f, (-1)^g\lambda^h) = (-1)^{eh+fg+fh}, \qquad e,f,g,h \in \{0,1\},$$

and is a symmetric bicharacter. For a prime p, each $y \in \mathcal{O}/(p)$ has $1 + \chi_p^3(y)$ square roots. Grouping by $y = x^2$ gives

$$\Gamma_{\mathrm{quad}}(p) = \frac{1}{|p|} \sum_{y \bmod p} \big(1 + \chi_p^3(y)\big) e(y/p) = \gamma_3(p),$$

since $\sum_{y \bmod p} e(y/p) = 0$. The Chinese remainder theorem extends this equality to squarefree n. Together with cubic reciprocity, it identifies the above $\mathcal{R}$ with the quotient $\chi_b(a)/\chi_a(b)$ for coprime primary a, b. Furthermore $\chi_c(4) = (-2/c)_3 = (c/(-2))_3$ is a character modulo 2. Consequently $G(c) = \overline{\chi_c(4)}\Gamma_{\mathrm{quad}}(c)$ factors through a fixed ray class group and satisfies (4.1). Moreover $\mathcal{R}(c,c) = \chi_c(-1)$. Indeed c^2 is a square, so $\Gamma_{\mathrm{quad}}(c^2) = \Gamma_{\mathrm{quad}}(1) = 1$, while for squarefree c the identity $\Gamma_{\mathrm{quad}}(c) = \gamma_3(c)$ and the inverse-character identity give $\Gamma_{\mathrm{quad}}(c)^2 = \gamma_3(c)^2 = \chi_c(-1)^3 = \chi_c(-1)$. Hence $\mathcal{R}(c,c) = \Gamma_{\mathrm{quad}}(c)^{-2} = \chi_c(-1)$. Both sides depend only on the class of c: the left side by the table above, and the right side because $\chi_c(-1) = (-1)^{(\mathrm{N}_{K/\mathbb{Q}}(c)-1)/6}$ depends only on $\mathrm{N}_{K/\mathbb{Q}}(c) \bmod 12$, hence only on $c \bmod 4$ for primary c. So this holds on the whole ray class group. The explicit formula also gives $G(1) = 1$ and $|G| = 1$ on every class.

Proof of (4.7). In the fixed ray class group, the multiplicative relation for G, together with $G(1) = 1$, $|G| = 1$ and $\mathcal{R}(a,a) = \chi_a(-1)$ from the proof of (4.1), gives

$$G(a^{-1}) = \chi_a(-1)\overline{G(a)}.$$

Using the symmetry and multiplicativity of $\mathcal{R}$, we obtain

$$G(ba^{-1}) = G(b)G(a^{-1})\mathcal{R}(b,a^{-1}) = \chi_a(-1)\overline{G(a)}G(b)\mathcal{R}(a,b),$$

which is (4.7).

Proof of (4.4)–(4.5). The second identity in (4.3) follows from the inverse-character Gauss identity: $\gamma_1(n)\gamma_{-1}(n) = \chi_n(-1)$. It remains to prove (4.4)–(4.5). Multiplying the second identity in (4.2) by $\overline{\alpha(n)}$ gives (4.4), and combining it with this inverse-character identity gives (4.5). □

A.2 The cubic theta transformation with fixed ray class twists

We sketch the proofs of Proposition 6.2, Lemma 6.3 and Lemma 6.4. The method is the theta transformation of Dunn and Radziwiłł [5, §5 and Appendix A], which extends Patterson [23] and Yoshimoto [28]. What is new compared with [5] is the fixed ray class twist Ψ_0 and the non-coprime part of the translation (the reduced denominator c_0 below). We record each step of the computation and indicate how it is verified; the formalization carries out every step in full (Appendix C).

The data are as in Section 6: Ψ_0 and S are fixed, $\mathcal{P}$ is a varying finite set of primes outside S, and $j_p \in \{0,\dots,5\}$ are the exponents of the character factors of $\Psi = \Psi_0 \prod_{p\in\mathcal{P}} \chi_p^{j_p}$. Write $w = (z,v) \in \mathbb{C}\times\mathbb{R}_{>0}$, and $\breve{e}(z) = \exp(2\pi\mathrm{i}(z+\bar{z}))$, so that $e(z) = \breve{e}(z/\lambda)$.

Step 1: finite Fourier expansion. Define $\phi : \mathcal{O} \to \mathbb{C}$ by $\phi(n) = \chi_n(\lambda)^2\Psi_0(n)$ if $n \equiv 1 \pmod 3$ and $(n,S) = 1$, and $\phi(n) = 0$ otherwise, and put

$$\Theta_\Psi(z,v) = \sum_{\substack{\ell\in\lambda^{-3}\mathcal{O}\\ \ell\neq 0}} \overline{\tau(-\ell)}\phi(\lambda^3\ell)\Big(\prod_{p\in\mathcal{P}} \chi_p^{j_p}(\lambda^3\ell)\Big) v K_{1/3}(4\pi|\ell|v)\breve{e}(\ell z). \tag{A.2}$$

For $\Psi = \Psi_k$ this is Θ_k from (6.2). Fix $L \in \mathcal{O}$ with all prime divisors in S, divisible by the conductor of Ψ_0 and by sufficiently high powers of 2 and λ; in particular $p \nmid L$ for $p \in \mathcal{P}$. By the supplementary law of cubic reciprocity [5, (1.5)], ϕ is periodic modulo L; let $\widehat{\phi}(h_0)$, $h_0 \in \mathcal{O}/(L)$, be its finite Fourier coefficients; since $|\phi| \le 1$, $|\widehat{\phi}(h_0)| \le 1$. Expanding each χ_p^j in additive characters modulo p,

$$\chi_p^j(x) = \sum_{h_p\in\mathcal{O}/(p)} C_{p,j}(h_p)e(h_p x/p), \qquad C_{p,j}(h_p) = \frac{1}{\mathrm{N}_{K/\mathbb{Q}}(p)} \sum_{y \bmod p} \chi_p^j(y)e(-h_p y/p), \tag{A.3}$$

and multiplying over $p \in \mathcal{P}$ writes Θ_Ψ as a finite combination of translates of $\overline{\theta}$:

$$\Theta_\Psi(z,v) = \sum_{\boldsymbol{h}} c_{\mathrm{F}}(\boldsymbol{h})\overline{\theta(z+z_{\boldsymbol{h}},v)}, \qquad z_{\boldsymbol{h}} = \lambda^2\Big(\frac{h_0}{L} + \sum_{p\in\mathcal{P}} \frac{h_p}{p}\Big), \qquad c_{\mathrm{F}}(\boldsymbol{h}) = \widehat{\phi}(h_0)\prod_{p\in\mathcal{P}} C_{p,j_p}(h_p). \tag{A.4}$$

The constant terms cancel, since $\sum_{\boldsymbol{h}} c_{\mathrm{F}}(\boldsymbol{h}) = \phi(0)\prod \chi_p^{j_p}(0) = 0$. Changing variables in the Gauss sums gives

$$C_{p,j}(h_p) = \begin{cases} \mathrm{N}_{K/\mathbb{Q}}(p)^{-1/2}\chi_p(-1)^j\gamma_j(p)\chi_p(h_p)^{-j}, & j \neq 0,\ h_p \neq 0,\\ 0, & j \neq 0,\ h_p = 0,\\ -\mathrm{N}_{K/\mathbb{Q}}(p)^{-1}, & j = 0,\ h_p \neq 0,\\ 1-\mathrm{N}_{K/\mathbb{Q}}(p)^{-1}, & j = 0,\ h_p = 0. \end{cases} \tag{A.5}$$

Call p *active* if $h_p \neq 0$. By (A.5), every p with $j_p \neq 0$ is active in a nonzero term. At an active prime, $v_p(\lambda^2 h_p/p) = -1$ while the other summands of $z_{\boldsymbol{h}}$ are p-integral, so the reduced denominator of $z_{\boldsymbol{h}}$ is $c = c_0 r$ up to a unit, with $r = \prod_{p \text{ active}} p$ and c_0 the reduced denominator of $\lambda^2 h_0/L$. This holds also when $(h_0, L) \neq 1$. There are $O_L(2^{|\mathcal{P}|})$ groups of translates, indexed by h_0 and the active set.

Step 2: the cusp and the multiplier. Write $z_{\boldsymbol{h}} = a/c$ in lowest terms, with $a \equiv 1 \pmod 3$ if $\lambda \mid c$ and $c \equiv 1 \pmod 3$ otherwise. Choose $g = \left(\begin{smallmatrix} a & b_g\\ c & \delta'\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathcal{O})$ with δ' prescribed modulo a fixed power of L and modulo r; this is possible by the Chinese remainder theorem, since $(a,c) = 1$. Then factor $g = g_1 H$, where $g_1 \equiv I \pmod 3$ and H is

$$H = \begin{cases} I, & 3 \mid c,\\ \left(\begin{smallmatrix} 1 & 0\\ u_0 & 1\end{smallmatrix}\right), & v_\lambda(c) = 1, \quad u_0 \in \{\lambda, -\lambda\}, \quad u_0 \equiv c \pmod 3,\\ \left(\begin{smallmatrix} u_0 & -1\\ 1 & 0\end{smallmatrix}\right), & (c,\lambda) = 1, \quad u_0 \equiv a \pmod 3, \end{cases}$$

as in [5, §5.1], with u_0 from a fixed set of representatives modulo 3 in the last case.[4] The invariance of θ under $\mathrm{SL}_2(\mathbb{Z})$ and under translations by $\mathbb{Z}+3\mathcal{O}$ reduces $\overline{\theta}(Hw)$ to one of the three cusp expansions (6.8), with representatives $\gamma_0, \gamma_+, \gamma_-$ (numbered $\gamma_1, \gamma_{10}, \gamma_{19}$ in [5, §5.1] and [23, Table II]). Writing $t_\sigma(\ell)$ for the coefficient of $vK_{1/3}(4\pi|\ell|v)\check{e}(\ell z)$ in $\theta(\gamma_\sigma w)$, Patterson's cusp calculation [23, Theorem 8.1 and Table III], in the form of [5, (5.13), (5.14)], gives

$$t_0(\ell)=\tau(\ell), \qquad t_-(\ell)=\omega^2\tau_1(\omega^2\ell)\check{e}(\ell), \qquad t_+(\ell)=\omega\tau_2(\omega\ell)\check{e}(\ell), \tag{A.6}$$

and, since complex conjugation sends the frequency ℓ to $-\ell$,

$$d_\sigma(\ell)=\overline{t_\sigma(-\ell)}, \qquad \sigma\in\{0,+,-\}. \tag{A.7}$$

The explicit formulas for τ, τ_1, τ_2 give the support (6.10). For $\ell = u\lambda^m n b^3$ they give $|t_0(\ell)| \le 3^{k/2+2}|b|$ if $m = 3k-4$ with $k \ge 1$, $|t_0(\ell)| \le 3^{k/2+5/2}|b|$ if $m = 3k-3$ with $k \ge 0$, and $|t_\pm(\ell)| \le 9|b|$ at $m=-4$. This gives the bound (6.11). For $d_\pm$ the coefficients are supported on $v_\lambda(\ell) = -4$, i.e. $m=-4$, so the bound is needed only there. This proves the coefficient assertions of Lemma 6.4.

The automorphy law of the cubic theta function is

$$\theta(g_1 w)=\kappa(g_1)\theta(w), \qquad \kappa(g_1)=(c_1/a_1)_3, \qquad g_1=\left(\begin{smallmatrix} a_1 & b_1 \\ c_1 & d_1\end{smallmatrix}\right)\equiv I \pmod 3, \tag{A.8}$$

where κ is Kubota's cubic character in the normalization of [5, (5.4), (5.6)].[5] Using the determinant equation and cubic reciprocity, one finds

$$\kappa(g_1)=\kappa_0\,(a/r)_3=\kappa_0\prod_{p\mid r}\chi_p(a)^2, \qquad |\kappa_0|=1, \tag{A.9}$$

where κ_0 depends only on the residues at the primes in S. These are fixed once h_0, the active set, and r modulo a fixed power of L are fixed. In the coordinates $(z',v')=g^{-1}(z+a/c,v)$, the Fourier mode of index ℓ at $z=0$ acquires the phase $\check{e}(-\delta'\ell/c)$. By the additive Chinese remainder theorem this phase splits as

$$\check{e}(-\delta'\ell/c)=\psi(\lambda^4\ell)\prod_{p\mid r}e(\epsilon_p h_p^{-1}\lambda^4\ell/p), \tag{A.10}$$

where ψ ranges over a fixed finite family of additive characters and $\epsilon_p\in(\mathcal{O}/p)^\times$ is determined by c and p.

Step 3: the local transformation. For each active prime, combining (A.5), the conjugate of (A.9), and (A.10), and summing over $h_p\neq 0$, gives

$$\sum_{h_p\neq 0}C_{p,j}(h_p)\chi_p(a)^{-2}e(\epsilon_p h_p^{-1}\lambda^4\ell/p)=\chi_p(\sigma_p)^{-2}\omega_{p,j}B_{p,j}(\lambda^4\ell), \qquad |\omega_{p,j}|=1, \tag{A.11}$$

with $\sigma_p=\lambda^2 c/p$ and $B_{p,j}$ from (6.5). Indeed, $a\equiv\sigma_p h_p \pmod p$, so $\chi_p(h_p)^{-j}\chi_p(a)^{-2}=\chi_p(\sigma_p)^{-2}\chi_p(h_p)^{-j-2}$, and the sum over $y=h_p^{-1}$ is a Gauss sum of χ_p^{j+2}. For $j\neq 0,4$ this character is nontrivial and the sum is $\mathrm{N}_{K/\mathbb{Q}}(p)^{1/2}$ times a unit times $\chi_p(\lambda^4\ell)^{-j-2}$. For $j=4$ the character is trivial on nonzero residues and the sum is $-1+\mathrm{N}_{K/\mathbb{Q}}(p)\mathbf{1}_{p\mid\lambda^4\ell}$. For active $j=0$ the coefficient $-\mathrm{N}_{K/\mathbb{Q}}(p)^{-1}$ combines with a cubic Gauss sum. An inactive prime contributes

[4] One checks $g_1\equiv I \pmod 3$ in each case from these congruences on u_0 and the congruences imposed on δ', which give $\delta'\equiv 1$, $b_g\equiv 0 \pmod 3$ when $\lambda\mid c$ and $\delta'\equiv 0$, $b_g\equiv -1 \pmod 3$ when $(c,\lambda)=1$.

[5] The convention matters. The exponent of χ_p produced by $\overline{\kappa}$ below is -2; it combines with the exponent -1 from the Fourier coefficient (A.5) to give the quadratic character $\chi_p^{-3}=\chi_p^3$. With the other common convention $(c_1/d_1)_3$ the exponent -2 from $\overline{\kappa}$ would become $+2$, the total exponent would be $+1$, and the dual twist would be the sextic character χ_p rather than a quadratic one.

only the scalar $1-\mathrm{N}_{K/\mathbb{Q}}(p)^{-1}$. For $j=1$ we obtain $B_{p,1}(x)=\chi_p(x)^{-3}=\chi_p(x)^3$, the quadratic factor used in (6.18).

Step 4: the archimedean transform. For $\operatorname{Re} s>1$ let $\mathcal{T}(s,\Psi)$ be the Dirichlet series of the completed sum,

$$\mathcal{T}(s,\Psi)=\sideset{}{^*}\sum_{\substack{n\equiv 1\ (3)\\(n,S)=1}}\overline{\alpha(n)}\gamma_2(n)\Psi(n)\mathrm{N}_{K/\mathbb{Q}}(n)^{-s}\sum_{\substack{b\equiv 1\ (3)\\(b,S)=1}}\overline{\alpha(b)}^{\,3}\Psi(b)^3\mathrm{N}_{K/\mathbb{Q}}(b)^{-3s+1/2},$$

so that, by Mellin inversion,

$$T(X;\Psi)=\frac{1}{2\pi\mathrm{i}}\int_{(\sigma)}\widehat{V}_*(s-\tfrac12)\mathcal{T}(s,\Psi)X^{s-1/2}\,ds,\qquad \sigma>1. \tag{A.12}$$

Differentiating in $\bar z$ supplies a factor $\bar\ell$ in each Fourier mode. Together with the Bessel integral $\int_0^\infty v^{2s}K_{1/3}(4\pi|\ell|v)\,dv=2^{2s-1}\Gamma(s+1/3)\Gamma(s+2/3)(4\pi|\ell|)^{-2s-1}$ [21, (10.43.19)] and the coefficient formula (2.9), this gives

$$\mathcal{J}(s):=\int_0^\infty \partial_{\bar z}\Theta_\Psi(z,v)\Big|_{z=0}v^{2s-1}\,dv=\frac{3^{5/2}}{4}\Big(\frac{27}{(2\pi)^2}\Big)^s\Gamma(s+1/3)\Gamma(s+2/3)\mathcal{T}(s,\Psi). \tag{A.13}$$

The integral defining $\mathcal{J}(s)$ converges for every s. The defining Fourier series controls $v\to\infty$. As $v\to 0$, the automorphy (A.8) moves each translate to a cusp expansion at height $1/(\mathrm{N}_{K/\mathbb{Q}}(c)v)\to\infty$, where the horizontal derivative removes the constant term. So $\mathcal{J}$ is entire, and so is $\mathcal{T}(s,\Psi)$. Substituting $v\mapsto(\mathrm{N}_{K/\mathbb{Q}}(c)v)^{-1}$ in each translate gives the functional equation

$$\mathcal{J}(s)=-\sum_{\boldsymbol{h}}c_{\mathrm{F}}(\boldsymbol{h})\overline{\kappa(g_{1,\boldsymbol{h}})}\,\overline{\alpha(c_{\boldsymbol{h}})}^{\,2}\mathrm{N}_{K/\mathbb{Q}}(c_{\boldsymbol{h}})^{1-2s}\mathcal{J}^\vee_{\boldsymbol{h}}(1-s), \tag{A.14}$$

where $\mathcal{J}^\vee_{\boldsymbol{h}}(s)=\int_0^\infty\partial_z\{\overline{\theta(H_{\boldsymbol{h}}(z,v))}\}\big|_{z=-\delta'_{\boldsymbol{h}}/c_{\boldsymbol{h}}}v^{2s-1}\,dv$ is the corresponding Mellin transform of the cusp expansion; by the cusp coefficients and the Bessel integral,

$$\mathcal{J}^\vee_{\boldsymbol{h}}(s)=\frac{\mathrm{i}\,\Gamma(s+1/3)\Gamma(s+2/3)}{4(2\pi)^{2s}}\sum_{0\neq\ell\in\lambda^{-4}\mathcal{O}}\frac{d_{\sigma_{\boldsymbol{h}}}(\ell)\alpha(\ell)}{\mathrm{N}_{K/\mathbb{Q}}(\ell)^s}\breve{e}(-\delta'_{\boldsymbol{h}}\ell/c_{\boldsymbol{h}}). \tag{A.15}$$

Both $\mathcal{T}(s,\Psi)$ and the series (A.15) at $1-s$ converge absolutely on the outer sides of the strip $0\le\operatorname{Re}s\le 1$. The functional equation and Stirling's formula give polynomial bounds there, and inside the strip by Phragmén–Lindelöf (compare [5, Propositions 5.1–5.2]).[6] This justifies moving the contour in (A.12) to $\operatorname{Re}s<0$. No pole is crossed, since $\mathcal{T}$ is entire. Now insert (A.15) into (A.14), sum over the nonzero h_p with (A.11), and set $t=\frac12-s$. This produces the gamma quotient and the weight $V_*^\sharp$ of (6.6), evaluated at $\mathrm{N}_{K/\mathbb{Q}}(\ell)X/\mathrm{N}_{K/\mathbb{Q}}(c)^2$, and the scalar

$$C=-\frac{\mathrm{i}}{81}\overline{\alpha(c)}^{\,2}\widehat{\phi}(h_0)\overline{\kappa_0}\prod_{p\text{ inactive}}(1-\mathrm{N}_{K/\mathbb{Q}}(p)^{-1})\prod_{p\text{ active}}\chi_p(\sigma_p)^{-2}\omega_{p,j_p},\qquad |C|\le 1/81.$$

The numerator gamma factors have their first pole at $t=-5/6$, so the kernel contour can be moved to $\operatorname{Re}t=0$. This proves Proposition 6.2.

Uniformity and weight decay. For the family Ψ_{k_0} of Lemma 6.3, every prime dividing k_0 has exponent $j_p=1$ and is active. So $r=k_0\prod_{p\in\mathcal{A}_{\mathrm{fix}}}p$, and fixing k_0 modulo a fixed power of L fixes H, c_0, κ_0 and the residues in (A.10). Hence d, ψ and c_0 depend on k_0 only through that

[6] The formalization uses a Gaussian-regularized contour shift, which needs only exponential growth in the strip.

residue class, which is Lemma 6.3. Finally, shifting the kernel contour in (6.6) to $\operatorname{Re} t = -1/4$ for $x \le 1$ and to $\operatorname{Re} t = A$ for $x \ge 1$, and using Stirling's formula, gives

$$|(x\partial_x)^j V_*^\sharp(x)| \ll_{A,j} \min\{x^{1/4}, (1+x)^{-A}\} \sup_{-A\le\eta\le 1/4} \int_{\mathbb{R}} (1+|u|)^{\lceil 4A\rceil + j + 2} |\widehat{V}_*(\eta + \mathrm{i}u)|\, du. \quad \text{(A.16)}$$

Since V_* is supported in a fixed compact interval, repeated integration by parts bounds the integral by $\|V_*\|_{C^J(I)}$ for some $J = J(A,j)$. This proves (6.12) and completes the proof of Lemma 6.4.

B Separating variables in smooth weights

This appendix justifies the separation of smooth weights in the completed mean-square estimate of Section 6, and the treatment of kernels depending on the row in the Poisson reductions of Sections 4 and 7.

B.1 Separating the variables

We use Mellin inversion to separate the variables of a smooth weight, with coefficient bounds controlled by finitely many derivatives.

Lemma B.1. *Fix a box $\mathcal{I} = I_1 \times \cdots \times I_d$, where each I_j is a compact interval in $(0,\infty)$. Every $\mathcal{K} \in C_c^\infty((0,\infty)^d)$ supported in $\mathcal{I}$ has a representation*

$$\mathcal{K}(\mathbf{x}) = \int_{\mathbb{R}^d} b(\mathbf{t}) \prod_{j=1}^d x_j^{\mathrm{i}t_j}\, d\mathbf{t}, \qquad x_j > 0. \quad \text{(B.1)}$$

For $J \ge 0$ and every even integer $q > J + d$, the coefficient satisfies

$$\int_{\mathbb{R}^d} |b(\mathbf{t})|(1+|\mathbf{t}|)^J\, d\mathbf{t} \ll_{\mathcal{I},J,q,d} \|\mathcal{K}\|_{C^q(\mathcal{I})}.$$

Proof. Take the Fourier transform in logarithmic coordinates:

$$b(\mathbf{t}) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{K}(e^{y_1}, \dots, e^{y_d}) e^{-\mathrm{i}\mathbf{t}\cdot\mathbf{y}}\, d\mathbf{y}.$$

Applying Fourier inversion to this gives (B.1). Since the intervals I_j are fixed and bounded away from zero, the C^q norm in logarithmic coordinates is bounded by a constant times $\|\mathcal{K}\|_{C^q(\mathcal{I})}$. Applying $(1-\Delta_{\mathbf{y}})^{q/2}$ under the integral therefore gives

$$|b(\mathbf{t})| \ll_{\mathcal{I},q,d} (1+|\mathbf{t}|)^{-q} \|\mathcal{K}\|_{C^q(\mathcal{I})} \qquad (q \ge 0 \text{ even}).$$

Multiplication by $(1+|\mathbf{t}|)^J$ and integration proves the bound when $q > J + d$. See also the multivariable Mellin formulation in [24, §10.1, (130)–(131)]. □

In our applications, the weight has the more specific form

$$\mathcal{K}_R(\mathbf{x}) = \prod_{j=1}^d W_j(x_j) F\Big(R \prod_{j=1}^d x_j^{a_j}\Big), \qquad R > 0,$$

where the exponents a_j are fixed real numbers and $W_j \in C_c^\infty((0,\infty))$ is supported in I_j. Assume that $F \in C^\infty((0,\infty))$ satisfies

$$\|F\|_{A,q} := \max_{0\le m\le q} \sup_{u>0} (1+u)^A \big|(u\partial_u)^m F(u)\big| < \infty \qquad (A \ge 0,\ q \in \mathbb{Z}_{\ge 0}).$$

On the fixed box $\mathcal{I}$, the argument of F is comparable to R. The chain rule and the pointwise bound in the proof give coefficients b_R satisfying, for $A \geq 0$ and even $q \geq 0$,

$$|b_R(\mathbf{t})| \ll (1+R)^{-A}(1+|\mathbf{t}|)^{-q}\|F\|_{A,q}\prod_{j=1}^{d}\|W_j\|_{C^q(I_j)}. \tag{B.2}$$

Consequently, for $J \geq 0$ and even $q > J + d$,

$$\int_{\mathbb{R}^d}|b_R(\mathbf{t})|(1+|\mathbf{t}|)^J\,d\mathbf{t} \ll (1+R)^{-A}\|F\|_{A,q}\prod_{j=1}^{d}\|W_j\|_{C^q(I_j)}. \tag{B.3}$$

The constants depend only on A, J, q, d, the intervals I_j, and the exponents a_j. Thus separation preserves the decay in R, and its cost is controlled by finitely many derivatives of the original weights.

B.2 Weights depending on the row

The next lemma extends mean-square bounds for a common test function to smooth kernels depending on the row, with losses controlled by uniform bounds on their derivatives.

Fix compact intervals $I \subset \operatorname{int} I_*$ in $(0,\infty)$ and an integer $m \geq 0$.

Lemma B.2. *Consider two families of finite sums*

$$S_{j,r}(U) = \sum_n a_{j,r}(n)U(x_{j,r,n}), \qquad j = 1,2, \quad x_{j,r,n} > 0,$$

with a finite set of rows r and nonnegative weights w_r. Suppose that, for every $U \in C_c^\infty(I_*)$,

$$\sum_r w_r|S_{j,r}(U)|^2 \leq M_j\|U\|^2_{C^m(I_*)}, \qquad j = 1,2.$$

Then for any $|c_r| \leq w_r$ and smooth kernels $\mathcal{K}_r$ supported in I^2,

$$\Big|\sum_r c_r\sum_{n_1,n_2}a_{1,r}(n_1)\overline{a_{2,r}(n_2)}\mathcal{K}_r(x_{1,r,n_1},x_{2,r,n_2})\Big| \ll_{I,I_*,m} \sqrt{M_1M_2}\sup_r\|\mathcal{K}_r\|_{C^{2m+4}(I^2)}. \tag{B.4}$$

Proof. Choose a real $V \in C_c^\infty(I_*)$ equal to one on I, and set $U_t(x) = V(x)x^{\mathrm{i}t}$. Apply (B.1) of Lemma B.1 to each $\mathcal{K}_r$ with $d = 2$ and $\mathcal{I} = I^2$. Replacing the second Mellin variable by its negative and multiplying by $V(x)V(y)$, which equals one on the support of $\mathcal{K}_r$, gives

$$\mathcal{K}_r(x,y) = \int_{\mathbb{R}^2} b_r(s,t)U_s(x)\overline{U_t(y)}\,ds\,dt,$$

with the following uniform bound, obtained from the pointwise coefficient estimate in the proof of Lemma B.1 with $d = 2$ and $\mathcal{I} = I^2$:

$$\sup_r|b_r(s,t)| \ll_{I,q} \frac{\sup_r\|\mathcal{K}_r\|_{C^q(I^2)}}{(1+|s|+|t|)^q} \qquad (q \geq 0 \text{ even}).$$

The hypothesis applies to each U_t, and $\|U_t\|_{C^m(I_*)} \ll_{I,I_*,m} (1+|t|)^m$. Taking the common bound for b_r before integrating and applying Cauchy–Schwarz in r therefore bounds the absolute value in (B.4) by a constant depending on I, I_*, m, q times

$$\sqrt{M_1M_2}\sup_r\|\mathcal{K}_r\|_{C^q(I^2)}\int_{\mathbb{R}^2}\frac{(1+|s|)^m(1+|t|)^m}{(1+|s|+|t|)^q}\,ds\,dt. \tag{B.5}$$

Taking $q = 2m+4$ makes the integral converge and proves the claim. □

B.3 Recombining the separated sums

We use weighted Cauchy–Schwarz to pass from mean-square bounds for the separated sums to a bound for their weighted integral.

Lemma B.3. *Let r, ι range over finite sets, let $d \geq 1$, and let $w_r \geq 0$. Suppose the measurable coefficients $c_{\iota,r}$ satisfy*

$$|c_{\iota,r}(\mathbf{t})| \leq b_\iota(\mathbf{t}), \qquad M := \sum_\iota \int_{\mathbb{R}^d} b_\iota(\mathbf{t})\, d\mathbf{t} < \infty,$$

where $b_\iota : \mathbb{R}^d \to [0,\infty)$. For measurable $F_{\iota,\mathbf{t}}(r)$, whenever the right-hand side is finite, one has

$$\sum_r w_r \Big|\sum_\iota \int c_{\iota,r}(\mathbf{t}) F_{\iota,\mathbf{t}}(r)\, d\mathbf{t}\Big|^2 \leq M \sum_\iota \int b_\iota(\mathbf{t}) \sum_r w_r |F_{\iota,\mathbf{t}}(r)|^2\, d\mathbf{t}. \tag{B.6}$$

The same assertion holds for finite sums with the integrals omitted.

Proof. For each fixed r, weighted Cauchy–Schwarz gives

$$\Big|\sum_\iota \int c_{\iota,r}(\mathbf{t}) F_{\iota,\mathbf{t}}(r)\, d\mathbf{t}\Big|^2 \leq M \sum_\iota \int b_\iota(\mathbf{t}) |F_{\iota,\mathbf{t}}(r)|^2\, d\mathbf{t}.$$

Multiply by w_r and sum over r to obtain (B.6). The same argument with sums in place of integrals proves the finite version. □

C Correspondence with the formal proof

The Lean development is the library `QRH` (Lean 4.34.1 with Mathlib). The statement checked by Comparator is

```
QRH.riemannZeta_ne_zero_of_twentyThree_twentyFourths_lt_re :
        (23 / 24 :  ) < s.re  riemannZeta s  0,
```

and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The directory `QRH/Spine` states the steps of this note as Lean theorems in the notation used here: the scale is D, and $M \geq bD$ is a cutoff beyond the support of W. (Inside the inherited library the roles are reversed: there Z is the scale and $D \in \mathbb{N}$ the cutoff.) The main theorem is proved by composing these steps exactly as in Section 3, and Proposition 3.1 is proved as Proposition 4.5 applied to Proposition 5.1. Names below are in the namespace `QRH` unless they start with `OAI`.

This note	Lean
$A_u(D)$, $A^\varphi(D)$	`A W D M u`, `weightedSum W D M` (`Spine/Setup`)
Theorem 1.1	`riemannZeta_ne_zero_of_twentyThree_twentyFourths_lt_re`
Proposition 3.1	`meanSquare` (`Spine/MeanSquare`)
Proposition 3.2	`saving_of_meanSquare` (`Spine/Extraction`)
Proposition 3.3	`riemannZeta_ne_zero_of_saving` (`Spine/ZeroFree`)
Lemma 4.1	`gauss_jacobi`, `gauss_two_cube`, `mobius_absorption`
Lemma 4.2	`poisson_with_exclusions`
Proposition 4.5	`poissonReduction`
Proposition 5.1	`dualBound`, `descent_bound`, `descent_by_rank`
Proposition 5.2	`completed_mean_square`
Lemma 5.3	`cube_inversion`, `short_cubes`, `descent_step`
Proposition 5.4	`transfer`; Lemma 7.2: `second_poisson`
Proposition 6.2	`theta_reflection`, `thetaModels`
Lemma 6.4	`theta_cusp_bounds`
Lemma 6.5 (Input D)	`quadratic_large_sieve`
Input C, (2.9)	`theta_automorphy`, `theta_coefficients`
$E_\varphi(s)$, $P_S(s)$, (3.6)	`OAI.SmoothMobiusCorrection.eulerCorrection`, `OAI.CompactMellinBridge.finiteEulerFactor`

Bookkeeping. The formal proof of Sections 4–7 follows the same strategy: Poisson summation in u, completed sums, theta reflection, the quadratic large sieve, cube inversion, and the transfer by two Poisson summations. It differs from the text in four points of bookkeeping, recorded in Remarks 4.6 and 5.5:

- test functions are expanded in height twists $\mathrm{N}_{K/\mathbb{Q}}(\mathfrak{n})^{\mathrm{i}\theta}$ once and for all, rather than controlled by C^J-norms;
- excluded primes are carried in the dual mean square as a mask, rather than removed by Lemma 4.4;
- rows are weighted by a compactly supported smooth majorant, whose Fourier transform only decays rapidly, rather than by a weight with compactly supported Fourier transform;
- the descent uses a fixed cube cutoff and an explicit loss budget.

The library contains no `sorry`, `axiom`, `native_decide` or `implemented_by`.

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